14.1 Standard Data Commands

To enable the system to read and write battery information, the BQ28Z620 uses a series of 2-byte standard I^{2}C commands with a 7-bit device address of 0x55 (8 bits = 0xAA to write and 0xAB to read). Each standard command has an associated command code pair, as indicated in Table 14-1, *Standard Commands*. Each protocol has specific means to access the data at each command code. Data RAM is updated and read by the gauge once per second.

Table 14-1. Standard Commands

Name		Register Code (LSB/MSB)	Description
ManufacturerAccess/ControlStatus()	CNTL	0x00/0x01	Control Register (see below)
AtRate()	AR	0x02/0x03	Read/Write. The value is a signed integer with the negative value indicating a discharge current value. The default value is 0 and forces *AtRateTimeToEmpty()* to return 65,535.
AtRateTimeToEmpty()	ARTTE	0x04/0x05	This read-only function returns an unsigned integer value to predict remaining operating time based on battery discharge at the *AtRate()* value in minutes with a range of 0 to 65,534. A value of 65,535 indicates *AtRate()* = 0. The gas gauge updates the *AtRateTimeToEmpty()* within 1 s after the system sets the *AtRate()* value. The gas gauge updates these parameters every 1 s. The commands are used in NORMAL mode.
Temperature()	TEMP	0x06/0x07	This read-only function returns an unsigned integer value of temperature in units (0.1 k) measured by the gas gauge, and is used for the gauging algorithm. It reports either *InternalTemperature()* or external thermistor temperature, depending on the setting of the ***[TEMPS]*** bit in Pack configuration.
Voltage()	VOLT	0x08/0x09	This read-only function returns an unsigned integer value of the measured cell pack in mV with a range of 0 12000 mV.
BatteryStatus()	FLAGS	0x0A/0x0B	See the Flags register.
Current()	INSTCURR	0x0C/0x0D	This read-only function returns a signed integer value that is the instantaneous current flow through the sense resistor. The value is updated every 1 s. Units are mA.
RemainingCapacity()	RM	0x10/0x11	This read-only command returns the predicted remaining capacity, based on rate (per configured Load Select), temperature, present depth-of-discharge, and stored impedance. Values are reported in mAh.
FullChargeCapacity()	FCC	0x12/0x13	This read-only command returns the predicted capacity of the battery at full charge, based on rate (per configured Load Select), temperature, present depth-of-discharge, and stored impedance. Values are reported in mAh.
AverageCurrent	AI	0x14/0x15	This read-only function returns a signed integer value that is the average current flow through the sense resistor. The value is updated every 1 s. Units are mA.
AverageTimeToEmpty	TTE	0x16/0x17	Uses average current value with a time constant of 15 s for this method. A value of 65535 means the battery is not being discharged.
AverageTimeToFull	TTF	0x18/0x19	This read-only function returns a unsigned integer value, predicting time to reach full charge for the battery in units of minutes based on *AverageCurrent()*. The computation accounts for the taper current time extension from linear TTF computation based on a fixed *AverageCurrent()* rate of charge accumulation. A value of 65,535 indicates the battery is not being charged.
StandbyCurrent	SI	0x1A/0x1B	This read-only function returns a signed integer value of measured standby current through the sense resistor. The *StandbyCurrent()* is an adaptive measurement. Initially it will report the standby current programmed in initial standby, and after several seconds in standby mode will report the measured standby. The register value is updated every 1 s when measured current is above the deadband and is less than or equal to 2 × initial standby. The first and last values that meet these criteria are not averaged in, since they may not be stable values. To approximate to a 1-min time constant, each new value of *StandbyCurrent()* is computed by taking approximate 93% weight of the last standby current and approximate 7% of the current measured average current.
StandbyTimeToEmpty	STTE	0x1C/0x1D	This read-only function returns a unsigned integer value, predicting remaining battery life at standby rate of discharge in units of minutes. The computation uses Nominal Available Capacity (NAC) for the calculation. A value of 65,535 indicates the battery is not being discharged.
MaxLoadCurrent	MLI	0x1E/0x1F	This read-only function returns a signed integer value in units of mA, of maximum load conditions. The *MaxLoadCurrent()* is an adaptive measurement which is initially reported as the maximum load current programmed in initial Max Load Current register. If the measured current is ever greater than the initial Max Load Current then the *MaxLoadCurrent()* updates to the new current. *MaxLoadCurrent()* is reduced to the average of the previous value and initial Max Load Current whenever the battery is charged to full after a previous discharge to an SOC of less than 50%. This will prevent the reported value from maintaining an unusually high value.
MaxLoadTimeToEmpty	MLTTE	0x20/0x21	This read-only function returns a unsigned integer value, predicting remaining battery life at the maximum discharge load current rate in units of minutes. A value of 65,535 indicates that the battery is not being discharged.

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Table 14-1. Standard Commands (continued)

Name		Register Code (LSB/MSB)	Description
AveragePower	AP	0x22/0x23	This read-only function returns a signed integer value of average power during battery charging and discharging. It is negative during discharge and positive during charge. A value of 0 indicates that the battery is not being discharged. The value is reported in units of mW.
InternalTemperature	INT_TEMP	0x28/0x29	This read-only function returns an unsigned integer value of the measured internal temperature of the device in 0.1-k units measured by the gas gauge.
CycleCount	CC	0x2A/0x2B	This read-only function returns an unsigned integer value of the number of cycles the battery has experienced a discharge (range 0 to 65535). One cycle occurs when accumulated discharge greater than or equal to CC threshold.
RelativeStateOfCharge	SOC	0x2C/0x2D	This read-only function returns an unsigned integer value of the predicted remaining battery capacity expressed as percentage of *FullChargeCapacity()* with a range of 0% to 100%.
StateOfHealth	SOH	0x2E/0x2F	This read-only function returns an unsigned integer value expressed as a percentage of the ratio of predicted FCC (25C ***SoH Load Rate***) over the *DesignCapacity()*. The range is 0x00 to 0x64 for 0% to 100%, respectively.
ChargeVoltage	CV	0x30/0x31	Returns the desired charging voltage in mV to the charger
ChargeCurrent	CC	0x32/0x33	Returns the desired charging current in mA to the charger
DesignCapacity		0x3C/0x3D	In SEALED and UNSEALED access: This command returns the value stored in Design Capacity and is expressed in mAh. This is intended to be a theoretical or nominal capacity of a new pack, but should have no bearing on the operation of the gas gauge functionality.
MACSubcmd		0x3E/0x3F	MAC Data block command
MACData		0x40/0x5F	MAC Data block
MACDataSum		0x60	MAC Data block checksum
MACDataLen		0x61	MAC Data block length

14.1.1 0x00/01 ManufacturerAccess() andControlStatus()

A write to this register is identical to writing to *MACSubcmd()* . For a description of MAC subcommands, see Section 14.2.

A read on this register returns the Control bits.

This Control Register is an I^{2}C register, and the control bits are read back on register 0x00/0x01. These control bits are provided for backward compatibility and ease-of-use.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD	SEC1	SEC0	AUTH CALM	RSVD	RSVD	Check Sum Valid	RSVD	RSVD	RSVD	RSVD	RSVD	LDMD	R_DIS	VOK	QMax

RSVD (Bit 15): Reserved

SEC1, SEC0 (Bits 14, 13): SECURITY mode

0, 0 = Reserved

0, 1 = Full Access

1, 0 = Unsealed

1, 1 = Sealed

AUTHCALM (Bit 12): Automatic CALIBRATION mode

1 = Enabled

0 = Disabled

RSVD (Bits 11–10): Reserved

CheckSumValid (Bit 9): Checksum is valid.

1 = Flash Writes are enabled.

0 = Flash Writes are disabled due to low voltage or PF condition.

RSVD (Bit 8–7): Reserved

RSVD (Bits 6–4): Reserved

LDMD (Bit 3): LOAD mode

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1 = Constant oower

0 = Constant current

R_DIS (Bit 2): Resistance updates

1 = Disabled

0 = Enabled

VOK (Bit 1): Voltage OK for QMax updates

1 = Detected

0 = Not detected

QMax (Bit 0): QMax updates. This bit toggles after every QMax update.

14.1.2 0x02/03 AtRate()

This read/write word function sets the value used in calculating *AtRateTimeToFull()* and *AtRateTimeToEmpty()*.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Default	Unit
		SE	US	FA						
0x02/03	*AtRate()*	R/W			Word	I2	–32768	32767	0	mA

14.1.3 0x04/05 AtRateTimeToEmpty()

This read-word function returns the remaining time to fully discharge the battery stack.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit	Note
		SE	US	FA						
0x04/05	*AtRateTimeToEmpty()*	R			Word	U2	0	65535	min	65535 indicates not being charged

14.1.4 0x06/07 Temperature()

This read-word function returns the temperature in units 0.1 K.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x06/07	*Temperature()*	R			Word	U2	0	65535	0.1 K

14.1.5 0x08/09 Voltage()

This read-word function returns the sum of the measured cell voltages.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x08/09	*Voltage()*	R			Word	U2	0	65535	mV

14.1.6 0x0A/0B BatteryStatus()

This read-word function returns various battery status information.

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SBS Cmd	Name	Access			Protocol	Type	Min	Max	Note
		SE	US	FA					
0x0A	*BatteryStatus()*	R	R	R	Word	H2			Bit 3:0: EC3,EC2,EC1,EC0 Error Code 0x00 = OK 0x1 = Busy 0x2 = Reserved Command 0x3 = Unsupported Command 0x4 = AccessDenied 0x5 = Overflow/Underflow 0x6 = BadSize 0x7 = UnknownError Bit 4: FD—Fully Discharged 0 = Battery ok 1 = Battery fully depleted Bit 5: FC—Fully Charged 0 = Battery not fully charged 01 = Battery fully charged Bit 6: DSG—Discharging 0 = Battery is charging. 1 = Battery is discharging. Bit 7: INIT—Initialization 0 = Inactive 1 = Active Bit 8: RTA—Remaining Time Alarm 0 = Inactive 1 = Active
0x0B	*BatteryStatus()*	R	R	R	Word	H2			Bit 9: RCA—Remaining Capacity Alarm 0 = Inactive 1 = Active Bit 10: Reserved Undefined Bit 11: TDA—Terminate Discharge Alarm 0 = Inactive 1 = Active Bit 12: OTA—Overtemperature Alarm 0 = Inactive 1 = Active Bit 13: Reserved Undefined Bit 14: TCA—Terminate Charge Alarm 0 = Inactive 1 = Active Bit 15: OCA—Overcharged Alarm 0 = Inactive 1 = Active

14.1.7 0x0C/0D Current()

This read-word function returns the measured current from the coulomb counter.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x0C/0D	*Current()*	R			Word	I2	–32767	32768	mA

14.1.8 0x0E/0F MaxError()

This read-word function returns the expected margin of error, in %, in the state-of-charge calculation with a range of 1 to 100%.

SBS Cmd	Name	Access			Protocol	Type	Min	Max	Unit
		SE	US	FA					
0x0E/0F	*MaxError()*	R			Word	U1	0	100	%

Condition	Action
Full device reset	*MaxError()* = 100%
RA-table only updated	*MaxError()* = 5%
QMax only updated	*MaxError()* = 3%
RA-table and QMax updated	*MaxError()* = 1%
Each *CycleCount()* increment after last valid QMax update	*MaxError()* increment by 0.05%
The ***Configuration:Max Error Time Cycle Equivalent*** period passed since the last valid QMax update	*MaxError()* increment by 0.05%.

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14.1.9 0x10/11 RemainingCapacity()

This read-word function returns the predicted remaining battery capacity.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x10/11	*RemainingCapacity()*	R			Word	U2	0	65535	mAh

14.1.10 0x12/13 FullChargeCapacity()

This read-word function returns the predicted battery capacity when fully charged.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x12/13	*FullChargeCapacity()*	R			Word	U2	0	65535	mAh

14.1.11 0x14/15 AverageCurrent()

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x14/15	*AverageCurrent()*	R			Word	I2	–32767	32768	mA

14.1.12 0x16/17 AverageTimeToEmpty()

This read-word function returns the predicted remaining battery capacity based on *AverageCurrent()*.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit	Note
		SE	US	FA						
0x16/17	*AverageTimeToEmpty()*	R			Word	U2	0	65535	min	65535 = Battery is not being discharged.

14.1.13 0x18/19 AverageTimeToFull()

This read-word function returns the predicted time to full charge based on *AverageCurrent()*.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit	Note
		SE	US	FA						
0x18/19	*AverageTimeToFull()*	R			Word	U2	0	65535	min	65535 = Battery is not being discharged.

14.1.14 0x1A/1B StandbyCurrent()

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x1A/1B	*StandbyCurrent()*	R			Word	I2	–32767	32768	mA

14.1.15 0x1C/1D StandbyTimeToEmpty()

This read-word function returns the predicted remaining battery capacity based on the minimum load rate of discharge.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit	Note
		SE	US	FA						
0x1C/1D	*StandbyTimeToEmpty()*	R			Word	U2	0	65535	min	65535 = Battery is not being discharged.

14.1.16 0x1E/1F MaxLoadCurrent()

This read-word function returns the maximum load conditions based on adaptive measurements, using the values in the initial Max Load Current register and the measured Max Load Current.

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SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x1E/1F	*MaxLoadCurrent()*	R			Word	I2	–32767	32768	mA

14.1.17 0x20/21 MaxLoadTimeToEmpty()

This read-word function returns the predicted remaining battery capacity based on the maximum load rate of discharge.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit	Note
		SE	US	FA						
0x20/21	*MaxLoadTimeTimeToEmpty()*	R	R	R	Word	U2	0	65535	min	65535 = Battery is not being discharged.

14.1.18 0x22/23 AveragePower()

This read-word function returns the average power during battery charging or discharging. It is negative due to discharge and positive due to charge. A zero value indicates the battery is not being discharged.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x22/23	*AveragePower()*	R			Word	I2	–32767	32768	mW

14.1.19 0x28/29 InternalTemperature()

This read-word function returns the internal die temperature in units 0.1 K.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x28/29	*InternalTemperature()*	R			Word	U2	0	65535	0.1 K

14.1.20 0x2A/2B CycleCount()

This read-word function returns the number of discharge cycles the battery has experienced.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x2A/2B	*CycleCount()*	R	R/W	R/W	Word	U2	0	65535	cycles

14.1.21 0x2C/2D RelativeStateOfCharge()

This read-word function returns the predicted remaining battery capacity as a percentage of *FullChargeCapacity()*.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x2C/2D	*RelativeStateOfCharge()*	R			Word	U1	0	100	%

14.1.22 0x2E/2F State-of-Health (SOH)

This command returns the state-of-health (SOH) information of the battery in percentage of design capacity. It is a read-only command.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit
		SE	US	FA					
0x2E/2F	*State-of-Health()*	R			Word	U1	0	100	%

14.1.23 0x30/31 ChargingVoltage()

This read-word function returns the desired charging voltage.

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SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit	Note
		SE	US	FA						
0x30/31	*ChargingVoltage()*	R	R	R	Word	U2	0	65535	mV	65535 = Request maximum voltage

14.1.24 0x32/33 ChargingCurrent()

This read-word function returns the desired charging current.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Unit	Note
		SE	US	FA						
0x32/33	*ChargingCurrent()*	R	R	R	Word	U2	0	65535	mA	65535 = Request maximum current

14.1.25 0x3C/3D DesignCapacity()

This read-word function returns the theoretical maximum pack capacity.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Default	Unit
		SE	US	FA						
0x3C/3D	*DesignCapacity()*	R	R/W	R/W	Word	U2	0	65535	4400 6336	mAh

14.1.26 0x3E/3F MACSubcmd()

Writes to this command will set the MAC subcommand and is interchangeable with *ManufacturerAccess()*.This command is provided to allow for executing full MAC subcommands in a single transaction, as well as to provide read-back verification of subcommands with a *MACData()* response. For a description of MAC subcommands see Section 14.2.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Default	Unit
		SE	US	FA						
0x3E/3F	*MACSubcmd()*	R	R	R	Word	—	—	—	—	—

14.1.27 0x40/0x5F MACData()

This is the data block for *MACSubcmd()* or *MACSubcmd()* commands.

SBS Cmd	Name	Access			Proto-col	Type	Min	Max	Default	Unit
		SE	US	FA						
0x40/5F	*MACData ()*	R	R	R	Block	—	—	—	—	—

14.1.28 0x60 MACDataChecksum()

This is the checksum of the *MACSubcmd()* and *MACData()* bytes.

SBS Cmd	Name	Access			Protocol	Type	Min	Max	Default	Unit	Note
		SE	US	FA							
0x60	*MACDataCheckSum ()*	R	R	R	Byte	—	—	—	—	—	

The checksum is the 8-bit sum of the MSB and LSB of the command plus the (command length) bytes in the buffer. The final sum is the bitwise inversion of the result. Since the length is part of the checksum, the verification cannot take place until the length is written. The checksum and length must be written together as a word to be valid.

14.1.29 0x61 MACDataLength()

This is the length for *MACSubcmd()* and *MACData()*.

SBS Cmd	Name	Access			Protocol	Type	Min	Max	Default	Unit	Note
		SE	US	FA							
0x61	*MACDataLength()*	R	R	R	Byte	—	—	—	—	—	

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The length byte for all MAC commands will include the 2-byte command, the 1-byte checksum, the 1-byte length, and 1 to 32 bytes of data. This means the minimum length value is 5 for a valid block (no length or checksum is used for command only writes, so block of zero size is not necessary). For proper write command validation, the checksum and length must be written in order (word access triggered).

14.2 Manufacturer Access Control (MAC)

The Manufacturer Access Control (MAC) system provides method for accessing subcommands to read and write data, and to control the device state.

The subcommands are set by a valid word access to either the *ManufacturerAccess()* or *MACSubcmd()* commands. For subcommands without data arguments, setting the subcommand alone will trigger execution. For subcommands with data the *MACData()*, *MACDataChecksum()* and *MACDataLength()* are used to write the subcommand arguments and trigger execution of the subcommand, or to read and validate the response after setting the subcommand. These can be accessed either in a single I2C transaction or split into multiple transactions as long as the access starts with the lowest address and ends at the highest, without other unrelated commands interleaved.

For subcommands with a data response, it is recommended to read back the *MACSubcmd()* registers to confirm the *MACData()* information comes from the intended command. There is no read back verification for subcommands that execute without any data arguments.

Example: Send a MAC *Gauging()* to enable IT via *MACSubcmd().*

1. With Impedance Track disabled, send *Gauging()* (0x0021) to *MACSubcmd()*
   a. I^2C Write, start address = 0x3E (or 0x00). Data = 21 00 (data must be sent in Little Endian).
2. IT is enabled, *ManufacturingStatus()[GAUGE_EN]* = 1.

Example: Read *Chemical ID()* (0x0006) via *MACSubcmd().*

1. Send *Chemical ID()* to *MACSubcmd().*
   a. I^2C Write, start address = 0x3E (or 0x00). Data sent = 06 00 (data must be sent in Little Endian).
2. Read the result from *MACSubcmd()* and *MfgMACData()*.
   a. I^2C Read, start address = 0x3E length = 36 bytes. First 4 bytes of the response will be 06 00 10 12.
   b. The first two bytes "06 00" is the MAC command (for verification).
   c. The second two bytes "10 12" are the Chem ID in Little Endian. That is, 0x1210 for ChemID 1210.
   d. The last two bytes of the 36-byte block will be the checksum and length. The length in this case will be 6. The checksum is 0xFF – (sum of the first length – 2 bytes). The length and checksum are used to validate the block response.

It is recommended to send “command only” operations to *ManufacturerAccess()* on 0x00 and 0x01, as this works with legacy devices as well as newer devices and also can reset any legacy compatibility options in effect; whereas, some legacy devices use 0x3E and 0x3F for other purposes. The 0x3E and 0x3F can always be used for reads. When no legacy devices are expected to be present, 0x3E and 0x3F *MACSubcmd()* can safely be used to optimize command execution.

To support detection of legacy devices, a request for the MAC subcommands DEV and VERSION written to 0x00 and 0x01 will cause the *ControlStatus()* read to report 0xFFA5. This token indicates that the device supports MAC on the 0x3E–0x61 command range and to go there for the response. In addition, the *MACDataLengh()* minimum valid value is 5 to allow some devices to have legacy support triggers when writing other values from 0–4 to this command.

Table 14-2. MACSubcmd() Command List

Command	Function	Access	Format	Data Read on MACData()	Not Available in SEALED Mode	Type	Units
0x0001	DeviceType	R	Block	√	—	hex	—
0x0002	FirmwareVersion	R	Block	√	—	hex	—

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Table 14-2. MACSubcmd() Command List (continued)

Command	Function	Access	Format	Data Read on MACData()	Not Available in SEALED Mode	Type	Units
0x0003	HardwareVersion	R	Block	√	—	hex	—
0x0004	IFChecksum	R	Block	√	—	hex	—
0x0005	StaticDFSignature	R	Block	√	—	hex	—
0x0006	ChemID	R	Block	√	—	hex	—
0x0007	Prev_MacWrite	R	Block	√	—	hex	—
0x0008	StaticChemDFSignature	R	Block	√	—	hex	—
0x0009	AllDFSignature	R	Block	√	—	hex	—
0x0010	ShutdownMode	W	—	—	—	hex	—
0x011	SleepMode	W	—	—	—	hex	—
0x012	Reset	W	—	—	—	hex	—
0x013	Auto_CAL_MAC	W	—	—	—	hex	—
0x001F	ChargeFET	W	—	—	√	hex	—
0x0020	DischargeFET	W	—	—	√	hex	—
0x0021	Gauging (IT Enable)	W	—	—	√	hex	—
0x0022	FETControl	W	—	—	√	hex	—
0x0023	LifetimeDataCollection	W	—	—	√	hex	—
0x0024	PermanentFailure	W	—	—	√	hex	—
0x0028	LifetimeDataReset	W	—	—	√	hex	—
0x0029	PermanentFailureDataReset	W	—	—	√	hex	—
0x002D	CalibrationMode	W	—	—	√	hex	—
0x002E	LifetimeDataFlush	W	—	—	√	hex	—
0x002F	LifetimeDataTest	W	—	—	√	hex	—
0x0030	SealDevice	W	—	—	—	hex	—
0x0035	SecurityKeys	R/W	Block	√	√	hex	—
0x0037	AuthenticationKey	W	Block	—	√	hex	—
0x0041	Reset	W	—	—	—	hex	—
0x004A	Device Name	R/W	Block	√	—	hex	—
0x004B	Device Chem	R/W	Block	√	—	hex	—
0x004C	Manufacturer Name	R/W	Block	√	—	hex	—
0x004D	Manufacturer Date	R/W	Block	√	—	hex	—
0x004E	Serial Number	R/W	—	√	—	hex	—
0x0050	SafetyAlert	R	Block	√	—	hex	—
0x0051	SafetyStatus	R	Block	√	—	hex	—
0x0052	PFAlert	R	Block	√	—	hex	—
0x0053	PFStatus	R	Block	√	—	hex	—
0x0054	OperationStatus	R	Block	√	—	hex	—
0x0055	ChargingStatus	R	Block	√	—	hex	—
0x0056	GaugingStatus	R	Block	√	—	hex	—
0x0057	ManufacturingStatus	R	Block	√	—	hex	—
0x0058	AFERegister	R	Block	√	—	hex	—
0x0060	LifetimeDataBlock1	R	Block	√	—	Mixed	Mixed
0x0070	ManufacturerData	R	Block	√	—	hex	—
0x0071	DAStatus1	R	Block	√	—	Mixed	Mixed
0x0072	DAStatus2	R	Block	√	—	Mixed	Mixed
0x0073	ITStatus1	R	Block	√	—	Mixed	Mixed
0x0074	ITStatus2	R	Block	√	—	Mixed	Mixed
0x0075	ITStatus3	R	Block	√	—	Mixed	Mixed
0x0076	CB Status	R	Block	√	—	hex	—
0x0077	FCC_SOH	R	Block	√	—	hex	—
0x01yy	DFAccessRowAddress	R/W	Block	—	√	hex	—
0x0F00	ROMMode	W	—	—	√	hex	—
0xF080	ExitCalibrationOutput	R/W	Block	√	√	hex	—
0xF081	OutputCCandADCforCalibration	R/W	Block	√	√	hex	—

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Table 14-2. MACSubcmd() Command List (continued)

Command	Function	Access	Format	Data Read on MACData()	Not Available in SEALED Mode	Type	Units
0xF082	OutputShortedCCandADC forCalibration	R/W	Block	√	√	hex	—

14.2.1 MACSubcmd() 0x0001 Device Type

The device can be checked for the IC part number.

Status	Condition	Action
Enable	0x0001 to *MACSubcmd()*	Returns the IC part number on subsequent read on *MACData()* in the following format: aaAA, where: aaAA: Device type

14.2.2 MACSubcmd() 0x0002 Firmware Version

The device can be checked for the firmware version of the IC.

Status	Condition	Action
Enable	0x0002 to *MACSubcmd()*	Returns the firmware revision on *MACData()* in the following format: ddDDvvVVbbBBTTzzZZRREE, where: ddDD: Device Number vvVV: Version bbBB: build number ttTT: Firmware type zzZZ: Impedance Track Version RR: Reserved EE: Reserved

14.2.3 MACSubcmd() 0x0003 Hardware Version

The device can be checked for the hardware version of the IC.

Status	Condition	Action
Enable	0x0003 to *MACSubcmd()*	Returns the hardware revision on subsequent read on *MACData()*

14.2.4 MACSubcmd() 0x0004 Instruction Flash Signature

The device can return the instruction flash signature

Status	Condition	Action
Enable	0x0004 to *MACSubcmd()*	Returns the IF signature on subsequent read on *MACData()* after a wait time of 250 ms

14.2.5 MACSubcmd() 0x0005 Static DF Signature

The device can return the data flash checksum.

Status	Condition	Action
Enable	0x0005 to *MACSubcmd()*	Returns the signature of all static DF on subsequent read on *MACData()* after a wait time of 250 ms. MSB is set to 1 if the calculated signature does not match the signature stored in DF.

14.2.6 MACSubcmd() 0x0006 Chemical ID

This command returns the chemical ID of the OCV tables used in the gauging algorithm.

Status	Condition	Action
Enable	0x0006 to *MACSubcmd()*	Returns the chemical ID on subsequent read on *MACData()*

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14.2.7 MACSubcmd() 0x0007 Pre_MACWrite

This command enables copying the last MAC into a 2-byte block.

Status	Condition	Action
Enable	0x0007 to *MACSubcmd()*	Copies the last MAC information into a 2-byte block *MACData()*

14.2.8 MACSubcmd() 0x0008 Static Chem DF Signature

The device can return the data flash checksum.

Status	Condition	Action
Enable	0x0008 to *MACSubcmd()*	Returns the signature of all static chemistry DF on a subsequent read on *MACData()* after a wait time of 250 ms. MSB is set to 1 if the calculated signature does not match the signature stored in DF.

14.2.9 MACSubcmd() 0x0009 All DF Signature

The device can return the data flash checksum.

Status	Condition	Action
Enable	0x0008 to *MACSubcmd()*	Returns the signature of all DF parameters on a subsequent read on *MACData()* after a wait time of 250 ms. MSB is set to 1 if the calculated signature does not match the signature stored in DF. It is normally expected that this signature will change due to update of lifetime, gauging, and other information.

14.2.10 MACSubcmd() 0x0010 SHUTDOWN Mode

The device can be sent to SHUTDOWN mode before shipping to reduce power consumption to a minimum. The device will wake up when a voltage is applied to PACK. When the pack is sealed, this feature requires the command be sent twice in a row for safety. Once SHUTDOWN mode is enabled, it is not possible to clear it without entering SHUTDOWN mode.

Status	Condition	Action
Normal	*OperationStatus()[SDM]* = 0	Shutdown feature is armed internally.
Arm	*OperationStatus()[SEC1,SEC0]* = [1, 1] 0x0010 sent to *MACSubcmd()*	Shutdown feature is armed internally.
Enable	0x0010 to *MACSubcmd()* when *OperationStatus[SEC1,SEC0]* = [1, 1] OR 0x0010 to *MACSubcmd()* when shutdown feature is armed.	*OperationStatus()[SDM]* = 1
Trip	*Current()* = 0 AND Voltage on PACK < Power. Charger Present AND *OperationStatus()[SDM]* = 1	No charging or discharging allowed. Device shutdown.
Recovery	Voltage on PACK terminal and device is in SHUTDOWN mode.	Device powers up (reset)

14.2.11 MACSubcmd() 0x0011 SLEEP Mode

The device can be sent to SLEEP with *MACSubcmd()* if the sleep conditions are met.

Status	Condition	Action
Enable	0x0011 to *MACSubcmd()*	*OperationStatus()[SLEEPM]* = 1
Activate	|*Current()*| < ***Power:Sleep Current***	Turn off DSG FET Turn off CHG FET if ***FET Options[SLEEPCHG]*** = 0 Device goes to sleep Device wakes up every Power ***:Sleep Voltage Time*** period to measure voltage and temperature Device wakes up every Power ***:Sleep Current Time*** period to measure current

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Status	Condition	Action
Exit	|*Current()*| > Configuration:Sleep Current	*OperationStatus()[SLEEPM]* = 0 Return to NORMAL mode
Exit	Wake Comparator trips	*OperationStatus()[SLEEPM]* = 0 Return to NORMAL mode
Exit	*SafetyAlert()* flag or *PFAlert()* flag set	*OperationStatus()[SLEEPM]* = 0 Return to NORMAL mode

14.2.12 MACSubcmd() 0x0012 Device Reset

This command resets the device.

Status	Condition	Action
Enable	0x0012 to *MACSubcmd()*	Resets the device

Note

Command 0x0041 also resets the device.

14.2.13 MACSubcmd() 0x001F CHG FET

This command turns on/off CHG FET drive function to ease testing during manufacturing.

Status	Condition	Action
Disable	*ManufacturingStatus()[FET,CHG]* = 0,1 AND 0x001F to *MACSubcmd()*	*ManufacturingStatus()[FET,CHG]* = 0,0 CHG FET turns off
Enable	*ManufacturingStatus()[FET,CHG]* = 0,0 AND 0x001F to *MACSubcmd()*	*ManufacturingStatus()[FET,CHG]* = 0,1 CHG FET turns on if no safety condition override.

14.2.14 MACSubcmd() 0x0020 DSG FET

This command turns on/off DSG FET drive function to ease testing during manufacturing.

Status	Condition	Action
Disable	*ManufacturingStatus()[FET,DSG]* = 0,1 AND 0x0020 to *MACSubcmd()*	*ManufacturingStatus()[FET,DSG]* = 0,0 DSG FET turns off *ManufacturingStatus()[FET,DSG]* = 0,0 AND 0x0020 to *MACSubcmd()*
Enable		*ManufacturingStatus()[FET,DSG]* = 0,1 DSG FET turns on if no safety condition override.

14.2.15 MACSubcmd() 0x0021 Gauging

This command enables or disables the gauging function to ease testing during manufacturing.

Status	Condition	Action
Disable	*ManufacturingStatus()[Gauge]* = 1 AND 0x0021 to *MACSubcmd()*	*ManufacturingStatus()[Gauge]* = 0 Disables gauging feature
Enable	*ManufacturingStatus()[Gauge]* = 0 AND 0x0021 to *MACSubcmd()*	*ManufacturingStatus()[Gauge]* = 1 Enable gauging feature

14.2.16 MACSubcmd() 0x0022 FET Control

This command disables/enables control of the CHG, DSG, and PCHG FETs by the firmware.

Status	Condition	Action
Disable	*ManufacturingStatus()[FET]* = 1 AND 0x0022 to *MACSubcmd()*	*ManufacturingStatus()[FET]* = 0 CHG and DSG FETs are disabled and remain OFF.

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Status	Condition	Action
Enable	*ManufacturingStatus()[FET]* = 0 AND 0x0022 to *MACSubcmd()*	*ManufacturingStatus()[FET]* = 1 CHG and DSG FETs are controlled by the firmware.

14.2.17 MACSubcmd() 0x0023 Lifetime Data Collection

This command disables/enables Lifetime data collection for ease of manufacturing.

Status	Condition	Action
Disable	*ManufacturingStatus()[LF]* = 1 AND 0x0023 to *MACSubcmd()*	*ManufacturingStatus()[LF]* = 0 Lifetime Data collection feature is disabled.
Enable	*ManufacturingStatus()[LF]* = 0 AND 0x0023 to *MACSubcmd()*	*ManufacturingStatus()[LF]* = 1 Lifetime Data collection feature is enabled.

14.2.18 MACSubcmd() 0x0024 Permanent Failure

This command disables/enables Permanent Failure for ease of manufacturing.

Status	Condition	Action
Disable	*ManufacturingStatus()[PF]* = 1 AND 0x0024 to *MACSubcmd()*	*ManufacturingStatus()[PF]* = 0 Permanent Failure feature is disabled.
Enable	*ManufacturingStatus()[PF]* = 0 AND 0x0024 to *MACSubcmd()*	*ManufacturingStatus()[PF]* = 1 Permanent Failure feature is enabled.

14.2.19 MACSubcmd() 0x0028 Lifetime Data Reset

This command resets Lifetime data in data flash for ease of manufacturing.

Status	Condition	Action
Reset	0x0028 to *MACSubcmd()*	Clears Lifetime Data in DF

14.2.20 MACSubcmd() 0x0029 Permanent Fail Data Reset

This command resets PF data in data flash for ease of manufacturing.

Status	Condition	Action
Reset	0x0029 to *MACSubcmd()*	Clears PF Data in DF

14.2.21 MACSubcmd() 0x002D CALIBRATION Mode

This command disables/enables entry into CALIBRATION mode. Status is indicated by the *ManufacturingStatus()[CAL_EN]* flag.

Status	Condition	Action
Disable	*ManufacturingStatus()[CAL_EN]* = 1 AND 0x002D to *MACSubcmd()*	*ManufacturingStatus()[CAL_EN]* = 0 Disables output of ADC and CC raw data on *ManufacturingData()*
Enable	*ManufacturingStatus()[CAL_EN]* = 0 AND 0x002D to *MACSubcmd()*	*ManufacturingStatus()[CAL_EN]* = 1 Enables output of ADC and CC raw data on *ManufacturingData()*, controllable with 0xF081 and 0xF082 on *MACSubcmd()*

14.2.22 MACSubcmd() 0x0030 Seal Device

This command seals the device for the field, disabling certain commands and access to DF.

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Status	Condition	Action
Sealed	*OperationStatus()[SEC1,SEC0]* = 0,1 or 1,0 AND 0x0030 to *MACSubcmd()*	*OperationStatus()[SEC1,SEC0]* = 1,1 Certain commands are not available. See Table 14-1 for details.

14.2.23 MACSubcmd() 0x0035 Security Keys

This is a read/write command that changes the Unseal and Full Access keys. To read the keys, send the *SecurityKeys()* command to either the *MACSubcmd()* 0x00 or 0x3E, followed by a read from *MACSubcmd()*.

To change the keys, the write operations must be sent through *MACSubcmd()* 0x3E with the *SecurityKeys()* followed by the keys. Each parameter entry must be sent in Little Endian.

Example:

Changing the Unseal key to 0x0123, 0x4567 and the Full Access key to 0x89AB, 0xCDEF:

Byte0: Unseal Key LSB
Byte1: Unseal Key MSB
Byte2: Full Access Key LSB
Byte3: Full Access Key MSB

Write block: command through *MACSubcmd()* starting address 0x3E, data block (hex) = [35 00 23 01 67 45 AB 89 EF CD].

Starting address 0x60, data block(hex) = [0A 0C] (checksum followed by length)

Checksum = 0x0A = ~(0x35 + 0x00 + 0x23 + 0x01 + 0x67 + 0x45 + 0xAB + 0x89 + 0xEF + 0xCD). The checksum is the 8-bit sum of the MSB and LSB of the command plus the (command length) bytes in the buffer. The final sum is the bitwise inversion of the result.

14.2.24 MACSubcmd() 0x0037 Authentication Key

This command enters a new authentication key into the device.

Status	Condition	Action
Initiate	*OperationStatus()[SEC1,SEC0]* = 0,1 AND 0x0037 to *MACSubcmd()*	*OperationStatus()[AUTH]* = 1 160-bit random number available at *MACData()*
Enter Key	Correct 128-bit Key written to *MACData()* in the format 0xAABBCCDDEEFFGGHHIIJJKKLLMMNNOOPP, where AA is LSB. In addition to this information, the checksum + length data block is required.	Wait time 250 ms *OperationStatus()[AUTH]* = 0 Device returns 160-bit HMAC digest at *MACData()* in the format: 0xAABBCCDDEEFFGGHHIIJJKKLLMMNNOOPPQQRRSSTTTT, where AA is LSB. The HMAC digest was calculated using a challenge of all zeroes + key. The result can be used to verify the key without allowing a plain text read back.

14.2.25 MACSubcmd() 0x0041 Device Reset

This command resets the device.

Status	Condition	Action
Enable	0x0041 to *MACSubcmd()*	Reset the device

Note

Command 0x0012 also resets the device.

14.2.26 MACSubcmd() 0x004A Device Name

This command returns the assigned pack name, as programmed in data flash.

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Status	Condition	Action
Enable	0x004A to *MACSubcmd()*	Returns the value stored in ***Device Name***

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
I2C Configuration	Data	Device Name	S21	—	—	bq28z620	ASCII	Assigned pack name

14.2.27 MACSubcmd() 0x004B Device Chem

This command returns the battery chemistry used in the pack, as programmed in data flash.

Status	Condition	Action
Enable	0x004B to *MACSubcmd()*	Returns the value stored in ***Device Chemistry***

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
I2C Configuration	Data	Device Chemistry	S5	—	—	LION	ASCII	The battery chemistry used in the pack. This parameter does not change with chemistry programming and must be updated manually.

14.2.28 MACSubcmd() 0x004C Manufacturer Name

This command returns the pack manufacturer name, as programmed in data flash.

Status	Condition	Action
Enable	0x004C to *MACSubcmd()*	Returns the value stored in ***Manufacturer Name***

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
I2C Configuration	Data	Manufacturer Name	S21	—	—	Texas Instruments	ASCII	Pack manufacturer name

14.2.29 MACSubcmd() 0x004D Manufacture Date

This command returns the manufacture date of the pack, as programmed in data flash.

Status	Condition	Action
Enable	0x004D to *MACSubcmd()*	Returns the value stored in ***Manufacture Date***

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
I2C Configuration	Data	Manufacture Date	U2	0	65535	1980-1-1	date	Manufacture date of the pack, which follows the format: Day + Month×32 + (Year–1980) × 512 Example: 10/27/2017 = 19291 (or 0x4B5B, binary split: 100101, 1010, 11011)

14.2.30 MACSubcmd() 0x004E Serial Number

This command returns the serial number of the pack, as programmed in data flash.

Status	Condition	Action
Enable	0x004E to *MACSubcmd()*	Returns the value stored in ***Serial Number***

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
I2C Configuration	Data	Serial Number	H2	0x0000	0xFFFF	0x0001	—	Serial number of the pack

14.2.31 MACSubcmd() 0x0050 SafetyAlert

This command returns the *SafetyAlert()* flags on *MACSubcmd() or MACData()*.

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Status	Condition	Action
Activate	0x0050 to *MACSubcmd()*	Output *SafetyAlert()* flags on *MACData()*

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RSVD	RSVD	RSVD	RSVD	UTD	UTC	RSVD	RSVD	RSVD	RSVD	CTOS	RSVD	PTOS	RSVD	RSVD	RSVD
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD	RSVD	OTD	OTC	RSVD	ASCD	RSVD	ASCC	RSVD	AOLD	RSVD	OCD	RSVD	OCC	COV	CUV

UTD (Bit 27): Undertemperature During Discharge

1 = Detected

0 = Not Detected

UTC (Bit 26): Undertemperature During Charge

1 = Detected

0 = Not Detected

0 = Not Detected

CTOS (Bit 21): Charge Timeout Suspend

1 = Detected

0 = Not Detected

PTOS (Bit 19): Precharge Timeout Suspend

1 = Detected

0 = Not Detected

OTD (Bit 13): Overtemperature During Discharge

1 = Detected

0 = Not Detected

OTC (Bit 12): Overtemperature During Charge

1 = Detected

0 = Not Detected

ASCD (Bit 10): Short-Circuit During Discharge

1 = Detected

0 = Not Detected

ASCC (Bit 8): Short-Circuit During Charge

1 = Detected

0 = Not Detected

AOLD (Bit 6): Overload During Discharge

1 = Detected

0 = Not Detected

OCD (Bit 4): Overcurrent During Discharge

1 = Detected

0 = Not Detected

OCC (Bit 2): Overcurrent During Charge

1 = Detected

0 = Not Detected

COV (Bit 1): Cell Overvoltage

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1 = Detected

0 = Not Detected

CUV (Bit 0): Cell Undervoltage

1 = Detected

0 = Not Detected

14.2.32 MACSubcmd() 0x0051 SafetyStatus

This command returns the *SafetyStatus()* flags on *MACData()*.

Status	Condition	Action
Activate	0x0051 to *MACSubcmd()*	Outputs *SafetyStatus()* flags on *MACData()*

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RSVD	RSVD	RSVD	RSVD	UTD	UTC	RSVD	RSVD	RSVD	RSVD	RSVD	CTO	RSVD	PTO	RSVD	RSVD
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD	RSVD	OTD	OTC	RSVD	ASCD	RSVD	ASCC	RSVD	AOLD	RSVD	OCD	RSVD	OCC	COV	CUV

UTD (Bit 27): Undertemperature During Discharge

1 = Detected

0 = Not Detected

UTC (Bit 26): Undertemperature During Charge

1 = Detected

0 = Not Detected

CTO (Bit 20): Charge Timeout

1 = Detected

0 = Not Detected

PTO (Bit 18): Precharge Timeout

1 = Detected

0 = Not Detected

OTD (Bit 13): Overtemperature During Discharge

1 = Detected

0 = Not Detected

OTC (Bit 12): Overtemperature During Charge

1 = Detected

0 = Not Detected

ASCD (Bit 10): Short-Circuit During Discharge

1 = Detected

0 = Not Detected

ASCC (Bit 8): Short-Circuit During Charge

1 = Detected

0 = Not Detected

AOLD (Bit 6): Overload During Discharge

1 = Detected

0 = Not Detected

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OCD (Bit 4): Overcurrent During Discharge

1 = Detected

0 = Not Detected

OCC (Bit 2): Overcurrent During Charge

1 = Detected

0 = Not Detected

COV (Bit 1): Cell Overvoltage

1 = Detected

0 = Not Detected

CUV (Bit 0): Cell Undervoltage

1 = Detected

0 = Not Detected

14.2.33 MACSubcmd() 0x0052 PFAlert

This command returns the *PFAlert()* flags on *MACData()*.

Status	Condition	Action
Activate	0x0052 to *MACSubcmd()*	Outputs *PFAlert()* flags on *MACData()*

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	DFETF	CFETF
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD	RSVD	RSVD	VIMA	VIMR	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	SOV	RSVD

RSVD (Bits 31–18): Reserved

DFETF (Bit 17): Discharge FET Failure

1 = Detected

0 = Not Detected

CFETF (Bit 16): Charge FET Failure

1 = Detected

0 = Not Detected

RSVD (Bits 15–13): Reserved

VIMA (Bit 12): Voltage Imbalance While Pack Is Active Failure

1 = Detected

0 = Not Detected

VIMR (Bit 11): Voltage Imbalance While Pack Is At Rest Failure

1 = Detected

0 = Not Detected

RSVD (Bits 10–2): Reserved

SOV (Bit 1): Safety Cell Overvoltage Failure

1 = Detected

0 = Not Detected

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RSVD (Bit 0): Reserved

14.2.34 MACSubcmd() 0x0053 PFStatus

This command returns the *PFStatus()* flags on *MACData()*.

Status	Condition	Action
Activate	0x0053 to *MACSubcmd()*	Outputs *PFStatus()* flags on *MACData()*

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	DFETF	CFETF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD	RSVD	RSVD	VIMA	VIMR	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	SOV	RSVD

RSVD (Bits 31–18): Reserved

DFETF (Bit 17): Discharge FET Failure

1 = Detected

0 = Not Detected

CFETF (Bit 16): Charge FET Failure

1 = Detected

0 = Not Detected

RSVD (Bits 15–13): Reserved

VIMR (Bit 12): Voltage Imbalance While Pack Is At Rest Failure

1 = Detected

0 = Not Detected

VIMA (Bit 11): Voltage Imbalance While Pack Is Active Failure

1 = Detected

0 = Not Detected

RSVD (Bits 10–2): Reserved

SOV (Bit 1): Safety Cell Overvoltage Failure

1 = Detected

0 = Not Detected

RSVD (Bit 0): Reserved

14.2.35 MACSubcmd() 0x0054 OperationStatus

This command returns the *OperationStatus()* flags on *MACData()*.

Status	Condition	Action
Activate	0x0054 to *MACSubcmd()*	Outputs *OperationStatus()* flags on *MACData()*

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

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RSVD	RSVD	EM SHUT	CB	SLP CC	SLP AD	SMBL CAL	INIT	SLEEP M	XL	CAL_OFF SET	CAL	AUTO CALM	AUTH	RSVD	SDM
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SLEEP	SEC1	SEC0	PF	SS	SDV	SEC1	SEC0	RSVD	RSVD	RSVD	RSVD	RSVD	CHG	DSG	RSVD

RSVD (Bits 31–30): Reserved

EMSHUT (Bit 29): Emergency FET Shutdown

1 = Active

0 = Inactive

CB (Bit 28): Cell Balancing

1 = Active

0 = Inactive

SLPCC (Bit 27): CC measurement in SLEEP mode

1 = Active

0 = Inactive

SLPAD (Bit 26): ADC measurement in SLEEP mode

1 = Active

0 = Inactive

SMBLCAL (Bit 25): Auto-offset calibration when bus low is detected.

1 = Active

0 = Inactive

INIT (Bit 24): Initialization after full reset

1 = Active

0 = Inactive

SLEEPM (Bit 23): SLEEP mode

1 = Active

0 = Inactive

XL (Bit 22): 400-kHz mode

1 = Active

0 = Inactive

CAL_OFFSET (Bit 21): Calibration Output (raw ***CC Offset*** data)

1 = Active when MAC *OutputShortedCCADCCal()* is sent and the raw shorted CC data for calibration is available.

0 = When the raw shorted CC data for calibration is not available.

CAL (Bit 20): Calibration Output (raw ADC and CC data)

1 = Active when either the MAC *OutputCCADCCal()* or *OutputShortedCCADCCal()* is sent and the raw CC and ADC data for calibration is available.

0 = When the raw CC and ADC data for calibration is not available.

AUTHCALM (Bit 19): Auto CC Offset Calibration by MAC *AutoCCOffset()*

1 = The gauge receives the MAC *AutoCCOffset()* and starts the auto CC Offset calibration.

0 = Clear when the calibration is completed.

AUTH (Bit 18): Authentication in progress

1 = Active

0 = Inactive

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RSVD (Bit 17): Reserved

SDM (Bit 16): SHUTDOWN triggered through a command

1 = Active

0 = Inactive

SLEEP (Bit 15): SLEEP mode conditions met

1 = Active

0 = Inactive

SEC1, SEC0 (Bits 14, 13): SECURITY mode

0, 0 = Reserved

0, 1 = Full Access

1, 0 = Unsealed

1, 1 = Sealed

PF (Bit 12): PERMANENT FAILURE mode status

1 = Active

0 = Inactive

SS (Bit 11): SAFETY mode status

1 = Active

0 = Inactive

SDV (Bit 10): SHUTDOWN triggered via low pack voltage

1 = Active

0 = Inactive

SEC1, SEC0 (Bits 9–8): SECURITY mode

0, 0 = Reserved

0, 1 = Full Access

1, 0 = Unsealed

1, 1 = Sealed

RSVD (Bits 7–3): Reserved

CHG (Bit 2): CHG FET status

1 = Active

0 = Inactive

DSG (Bit 1): DSG FET status

1 = Active

0 = Inactive

RSVD (Bit 0): Reserved

14.2.36 MACSubcmd() 0x0055 ChargingStatus

This command returns the *ChargingStatus()* flags on *MACData()*.

Status	Condition	Action
Activate	0x0055 to *MACSubcmd()*	Outputs *ChargingStatus()* flags on *MACData()*

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VCT	MCHG	SU	IN	HV	MV	LV	PV	RSVD	OT	HT	STH	RT	STL	LT	UT

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VCT (Bit 15): Charge Termination

1 = Active

0 = Inactive

MCHG (Bit 14): Maintenance Charge

1 = Active

0 = Inactive

SU (Bit 13): Charge Suspend

1 = Active

0 = Inactive

IN (Bit 12): Charge Inhibit

1 = Active

0 = Inactive

HV (Bit 11): High Voltage Region

1 = Active

0 = Inactive

MV (Bit 10): Mid Voltage Region

1 = Active

0 = Inactive

LV (Bit 9): Low Voltage Region

1 = Active

0 = Inactive

PV (Bit 8): Precharge Voltage Region

1 = Active

0 = Inactive

OT (Bit 6): Over Temperature Region

1 = Active

0 = Inactive

HT (Bit 5): High Temperature Region

1 = Active

0 = Inactive

STH (Bit 4): Standard Temperature High Region

1 = Active

0 = Inactive

RT (Bit 3): Room Temperature Region

1 = Active

0 = Inactive

STL (Bit 2): Standard Temperature Low Region

1 = Active

0 = Inactive

LT (Bit 1): Low Temperature Region

1 = Active

0 = Inactive

UT (Bit 0): Under Temperature Region

1 = Active

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0 = Inactive

14.2.37 MACSubcmd() 0x0056 GaugingStatus

This command returns the *GaugingStatus()* flags on *MACData()*.

Note

The Battery Management Studio (bqStudio) tool splits the returned value of *GaugingStatus()* into two separate registers: *GaugingStatus* = Lowest Byte, and *ITStatus* = 2nd and 3rd bytes, with the 4th byte reserved.

Status	Condition	Action
Activate	0x0056 to *MACSubcmd()*	Outputs *GaugingStatus()* flags on *MACData()*

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	OCV FR	LDMD	RX	QMax	VDQ
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NSFM	RSVD	SLPQ Max	QEN	VOK	RDIS	RSVD	REST	CF	DSG	EDV	BAL_ EN	TC	TD	FC	FD

OCVFR (Bit 20): Open Circuit Voltage in Flat Region (during RELAX)

1 = Detected

0 = Not Detected

LDMD (Bit 19): LOAD mode

1 = Constant Power

0 = Constant Current

RX (Bit 18): Resistance Update (Toggles after every resistance update)

QMax (Bit 17): QMax Update (Toggles after every QMax update)

VDQ (Bit 16): Discharge Qualified for Learning (based on RU flag)

NSFM (Bit 15): Negative Scale Factor Mode

1 = Negative Ra Scaling Factor Detected

0 = Negative Ra Scaling Factor Not Detected

SLPQMax (Bit 13): QMax Update During Sleep

1 = Active

0 = Inactive

QEN (Bit 12): Impedance Track Gauging (Ra and QMax updates are enabled.)

1 = Enabled

0 = Disabled

VOK (Bit 11): Voltage OK for QMax Update

1 = Detected

0 = Not Detected

RDIS (Bit 10): Resistance Updates

1 = Disabled

0 = Enabled

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REST (Bit 9): Rest

1 = OCV Reading Taken

0 = OCV Reading Not Taken or Not in Relax

1 = Detected

0 = Not Detected

CF (Bit 7): Condition Flag

1 = *MaxError()* > Max Error Limit (Condition Cycle is needed.)

0 = *MaxError()* < Max Error Limit (Condition Cycle is not needed.)

DSG (Bit 6): Discharge/Relax

1 = Charging Not Detected

0 = Charging Detected

EDV (Bit 5): End-of-Discharge Termination Voltage

1 = Termination voltage reached during discharge

0 = Termination voltage not reached or not in DISCHARGE mode

BAL_EN (Bit 4): Cell Balancing

1 = Cell balancing is possible if enabled.

0 = Cell balancing is not allowed.

TC (Bit 3): Terminate Charge

1 = Detected

0 = Not Detected

TD (Bit 2): Terminate Discharge

1 = Detected

0 = Not Detected

FC (Bit 1): Fully Charged

1 = Detected

0 = Not Detected

FD (Bit 0): Fully Discharged

1 = Detected

0 = Not Detected

14.2.38 MACSubcmd() 0x0057 ManufacturingStatus

This command returns the *ManufacturingStatus()* flags on *MACData()*.

Status	Condition	Action
Activate	0x0057 to *MACSubcmd()*	Outputs *ManufacturingStatus()* flags on *MACData()*

15	14	13	12	11	10	9	8
CAL_EN	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD

7	6	5	4	3	2	1	0
RSVD	PF_EN	LF_EN	FET_EN	GAUGE_EN	DSG_TEST	CHG_TEST	RSVD

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CAL_EN (Bit 15): CALIBRATION Mode

1 = Enabled

0 = Disabled

RSVD (Bits 14–7): Reserved

PF_EN (Bit 6): Permanent Failure

1 = Enabled

0 = Disabled

LF_EN (Bit 5): Lifetime Data Collection

1 = Enabled

0 = Disabled

FET_EN (Bit 4): All FET Action

1 = Enabled

0 = Disabled

GAUGE_EN (Bit 3): Gas Gauging

1 = Enabled

0 = Disabled

DSG_TEST (Bit 2): Discharge FET Test

1 = Discharge FET test activated

0 = Disabled

CHG_TEST (Bit 1): Charge FET Test

1 = Charge FET test activated

0 = Disabled

RSVD (Bit 0): Reserved

14.2.39 MACSubcmd() 0x0058 AFE Register

This command returns the *AFERegister()* values on *MACData()*. These are the AFE hardware registers and are intended for internal debug use only.

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Status	Condition	Action
Activate	0x0058 to *MACSubcmd()*	Output AFE Register values on *MACData()* in the following format: AABBCCDDEEFFGGHHIIJJKKLLMMNNOOPPQQRRSSTTUU where: AA: AFE Interrupt Status. AFE Hardware interrupt status (for example, wake time, push-button, and so on) BB: AFE FET Status. AFE FET status (for example, CHG FET, DSG FET, input, and so on) CC: AFE RXIN. AFE I/O port input status DD:AFE Latch Status. AFE protection latch status EE: AFE Interrupt Enable. AFE interrupt control settings FF: AFE Control. AFE FET control enable setting GG: AFE RXIEN. AFE I/O input enable settings HH: II: JJ: KK: AFE Cell Balance. AFE cell balancing enable settings and status LL: AFE ADC/CC Control. AFE ADC/CC Control settings MM: AFE ADC Mux. AFE ADC channel selections NN: OO: AFE Control. AFE control on various HW based features PP: AFE Timer Control. AFE comparator and timer control QQ: AFE Protection. AFE protection delay time control RR: AFE OCD. AFE OCD settings SS: AFE SCC. AFE SCC settings TT: AFE SCD1. AFE SCD1 settings UU: AFE SCD2. AFE SCD2 settings

14.2.40 MACSubcmd() 0x0060 Lifetime Data Block 1

This command returns the Lifetime data on *MACData()*.

Status	Condition	Action
Activate	0x0060 to *MACSubcmd()*	Outputs lifetime data values of Voltage, Current, Power, and Temperature on *MACData()*

14.2.41 MACSubcmd() 0x0070 ManufacturerInfo

This command returns *ManufacturerInfo* on *MACData()*.

Status	Condition	Action
Activate	0x0070 to *MACSubcmd()*	Outputs 32 bytes of ManufacturerInfo on *MACData()* in the following format: AABBCCDDEEFFGGHHIIJJKKLLMMNN OOPPQQRRSSTTUUVVWWXXVVZZ112233 445566

14.2.42 MACSubcmd() 0x0071 DAStatus1

This command returns the CellVoltages, PackVoltage, BatVoltage, CellCurrents, CellPowers, Power, and AveragePower on *MACData()*.

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Status	Condition	Action
Activate	0x0071 to *MACSubcmd()*	Outputs 32 bytes of data on *MACData()* in the following format: aaAAbbBBccCCddDDeeEEffFFggGGhhHHiiIIjjJJkkKKllLLmmMMnn NNooOOppPP where: AAaa: Cell Voltage 1 BBbb: Cell Voltage 2 CCcc: DDdd: EEee: BAT Voltage. Voltage at the VC2 (BAT) terminal FFff: PACK Voltage GGgg: Cell Current 1. Simultaneous current measured during Cell Voltage1 measurement HHhh: Cell Current 2. Simultaneous current measured during Cell Voltage2 measurement IIii: JJjj: KKkk: Cell Power 1. Calculated using Cell Voltage1 and Cell Current 1 data LLll: Cell Power 2. Calculated using Cell Voltage2 and Cell Current 2 data MMmm: NNnn: OOoo: Power calculated by *Voltage()* × *Current()* PPpp: Average Power. Calculated by *Voltage()* × *AverageCurrent()*

14.2.43 MACSubcmd() 0x0072 DAStatus2

This command returns the internal temp sensor, TS1*MACData()*.

Status	Condition	Action
Activate	0x0072 to *MACSubcmd()*	Outputs 14 bytes of temperature data values on *MacData()* in the following format: aaAAbbBBccCCddDDeeEEffFFggGG where: AAaa: Int Temperature BBbb: TS1 Temperature CCcc: DDdd: EEee: FFff: GGgg:

14.2.44 MACSubcmd() 0x0073 ITStatus1

This command instructs the device to return Impedance Track related gauging information on *MACData()*.

Status	Condition	Action
Activate	0x0073 to *MACSubcmd()*	Outputs 24 bytes of IT data values on *MACData()* in the following format: aaAAbbBBccCCddDDeeEEffFFggGGhhHHiiIIjjJJkkKKllLL where: AAaa: True Rem Q. True remaining capacity in mAh from IT simulation before any filtering or smoothing function. This value can be negative or higher than FCC. BBbb: True Rem E. True remaining energy in cWh from IT simulation before any filtering or smoothing function. This value can be negative or higher than FCC. CCcc: Initial Q. Initial capacity calculated from IT simulation DDdd: Initial E. Initial energy calculated from IT simulation EEee: TrueFullChgQ. True full charge capacity FFff: TrueFullChgE. True full charge energy GGgg: T_sim. Temperature during the last simulation run (in 0.1K) HHhh: T_ambient. Current estimated ambient temperature used by the IT algorithm for thermal modeling IIii: RaScale 0. Ra table scaling factor of Cell1 JJjj: RaScale 1. Ra table scaling factor of Cell2 KKkk: CompRes1. Last computed resistance for Cell1 LLll: CompRes2. Last computed resistance for Cell2

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14.2.45 MACSubcmd() 0x0074 ITStatus2

This command instructs the device to return Impedance Track related gauging information on *MACData()*.

Status	Condition	Action
Activate	0x0074 to *MACSubcmd()*	Outputs 24 bytes of IT data values on *MACData()* in the following format: AABBCCDDEEFFggGGhhHHiiIIjjJJkkKKllLLmmMMnnNNooOO where: AA: Pack Grid. Active pack grid point (only valid in discharge) BB: LStatus—Learned status of resistance table Bit 3 | Bit 2 | Bit 1| Bit 0 QMax | ITEN | CF1 | CF0 CF1, CF0: QMax Status 0,0 = Battery OK 0,1 = QMax is first updated in learning cycle. 1,0 = QMax and resistance table updated in learning cycle ITEN: IT enable 0 = IT disabled 1 = IT enabled QMax: QMax field updates 0 = QMax is not updated in the field. 1= QMax is updated in the field. CC: Cell Grid 1. Active grid point of Cell1 DD: Cell Grid 2. Active grid point of Cell2 EE: N/A. RSVD FF: N/A. RSVD HHhhGGgg: State Time. Time passed since last state change (Discharge, Charge, Rest) IIii: DOD0_1. Depth of discharge for Cell1 JJjj: DOD0_2. Depth of discharge for Cell2 KKkk: DOD0 Passed Q. Passed charge since DOD0 LLll: DOD0 Passed Energy. Passed energy since the last DOD0 update MMmm: DOD0 Time. Time passed since the last DOD0 update NNnn: DODEOC_1. Cell 1 DOD@EOC OOoo: DODEOC_2. Cell 2 DOD@EOC

14.2.46 MACSubcmd() 0x0075 ITStatus3

This command instructs the device to return Impedance Track related gauging information on *MACData()*.

Status	Condition	Action
Activate	0x0075 to *MACSubcmd()*	Outputs 20 bytes of IT data values on *MACData()* in the following format: aaAAbbBBccCCddDDeeEEffFFggGGhhHHiiIIjjJJ where: AAaa: QMax 1. QMax of Cell 1 BBbb: QMax 2. QMax of Cell 2 CCcc: QMaxDOD0_1. Cell 1 DOD for Qmax DDdd: QMaxDOD0_2. Cell 2 DOD for Qmax EEee: QMaxPassedQ. Passed charge since DOD for Qmax recorded (mAh) FFff: QMaxTime. Time since DOD for Qmax recorded (hour / 16 units) GGgg: Tk. Thermal model “k” HHhh: Ta. Thermal model “a” IIii: RawDOD0_1. Cell 1 raw DOD0 measurement JJjj: RawDOD0_2. Cell 2 raw DOD0 measurement

14.2.47 MACSubcmd() 0x0076 CB Status

This command returns the status of the cell being balanced.

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Status	Condition	Action
Enable	0x0076 to *MACSubcmd()*	Outputs 10 bytes of IT data values on *MACData()* in the following format: aaAAbbBBccCCddDDeeEE AAaa: CBTime1. Cell 1 balance time remaining BBbb: CBTime2. Cell 2 balance time remaining CCcc: CBDOD_1. Cell 1 DOD when balance calculated DDdd: CBDOD_2. Cell 2 DOD when balance calculated EEee: CBTotalDODChg. Total DOD charge when balance calculated

14.2.48 MACSubcmd() 0x0077 State-of-Health

This command returns the state-of-health percentage.

Status	Condition	Action
Enable	0x0077 to *MACSubcmd()*	Returns the state-of-health percentage on subsequent read on *MACData()*

14.2.49 MACSubcmd() 0x0F00 ROM Mode

This command sends the device into ROM mode in preparation for reprogramming.

Status	Condition	Action
ROM Mode	*OperationStatus()[SEC1,SEC0]* = 0,1 AND 0x0F00 to *MACSubcmd()*	Device goes to ROM mode ready for update. ROM command 0x08 will return to firmware mode. (Note: ROM commands are sent to address 0x16 using SMB protocol.)

14.2.50 Data Flash Access() 0x4000–0x5FFF

Accessing data flash (DF) is only supported by the *MACSubcmd()* by addressing the physical address.

To write to the DF, send the starting address, followed by the DF data block. The DF data block is the intended revised DF data to be updated to DF. The size of the DF data block ranges from 1 byte to 32 bytes. All individual data must be sent in Little Endian.

Write to DF example:

Assuming: data1 locates at address 0x4000 and data2 locates at address 0x4002.
Both data1 and data2 are U2 type.

To update data1 and data2, send a block write with command = 0x3E

block = starting address + DF data block

= 0x00 + 0x40 + data1_LowByte + data1_HighByte + data2_LowByte + data2_HighByte

To read the DF, send a block write to the *MACSubcmd(),* followed by the starting address, then send a block read to the *MACSubcmd()*. The return data contains the starting address followed by 32 bytes of DF data in Little Endian.

Read from DF example:

Taking the same assuming from the read DF example, to read DF:

a. Send the write block with command 0x3E, block = 0x00 + 0x40

b. Send the read block with command 0x3E

The returned block = a starting address + 32 bytes of DF data

= 0x00 + 0x40 + data1_LowByte + data1_HighByte + data2_LowByte + data2_HighByte.... data32_LowByte + data32_HighByte

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The gauge supports an auto-increment on the address during a DF read. This greatly reduces the time required to read out the entire DF. Continue with the read from the DF example. If another read block is sent with command 0x3E, the gauge returns another 32 bytes of DF data, starting with address 0x4020.

14.2.51 MACSubcmd() 0xF080 Exit Calibration Output Mode

This command stops the output of calibration data to the *MACData()* command.

Status	Condition	Action
Activate	*MACData()* = 1 AND 0xF080 to *MACSubcmd()*	Stops output of ADC or CC data on *MACData()*

14.2.52 MACSubcmd() 0xF081 Output CC and ADC for Calibration

This command instructs the device to output the raw values for calibration purposes on *MACData()*. All values are updated every 250 ms and the format of each value is 2's complement, MSB first.

Status	Condition	Action
Disable	*ManufacturingStatus()[CAL_EN]* = 1 AND 0xF080 to *MACSubcmd()*	*ManufacturingStatus()[CAL_EN]* = 0 Stops output of ADC and CC data on *MACData()*
Enable	0xF081 to *MACSubcmd()*	*ManufacturingStatus()[CAL_EN]* = 1 Outputs the raw CC and AD values on *MACData()* in the format of ZZYYaaAAbbBBccCCddDDeeEEffFF ggGGhhHHiiIIjjJJkkKK: ZZ: rolling 8-bit counter, increments when values are refreshed. YY: status, 1 when *MACSubcmd()* = 0xF081, 2 when *MACSubcmd()* = 0xF082 AAaa: Current (Coulomb Counter) BBaa: Cell Voltage 1 CCaa: Cell Voltage 2 DDaa: EEee: FFff:PACK Voltage GGgg: VC2 (BAT) Voltage HHhh: Cell Current 1 Iiii: Cell Current 2 JJjj: KKkk:

14.2.53 MACSubcmd() 0xF082 Output Shorted CC and ADC for Calibration

This command instructs the device to output the raw values for calibration purposes on *MACSubcmd() or MACData()*. All values are updated every 250 ms and the format of each value is 2's complement, MSB first. This mode includes an internal short on the coulomb counter inputs for measuring offset.

Status	Condition	Action
Disable	*ManufacturingStatus()[CAL_EN]* = 1 AND 0xF080 to *MACSubcmd()*	*ManufacturingStatus()[CAL_EN]* = 0 Stops output of ADC and CC data on *MACSubcmd() or MACData()*

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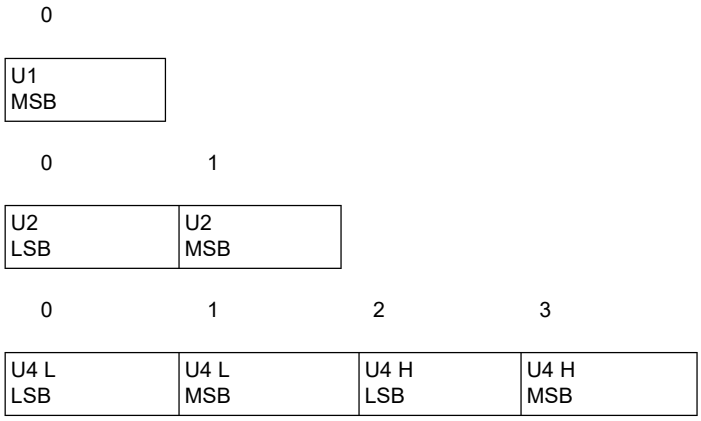
Status	Condition	Action
Enable	0xF082 to *MACSubcmd()*	*ManufacturingStatus()[CAL_EN]* = 1 Outputs the raw CC and AD values on *MACSubcmd() or MACData()* in the format of ZZYYaaAAbbBBccCCddDDeeEEffFF ggGGhhHHiiIIjjJJkkKK: ZZ: rolling 8-bit counter, increments when values are refreshed. YY: status, 1 when *MACSubcmd()* = 0xF081, 2 when *MACSubcmd()* = 0xF082 AAaa: Current (Coulomb Counter) BBaa: Cell Voltage 1 CCaa: Cell Voltage 2 DDaa: EEee: FFff:PACK Voltage GGgg: VC2 (BAT) Voltage HHhh: Cell Current 1 IIii: Cell Current 2 JJjj: KKkk:

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15.1 Data Formats

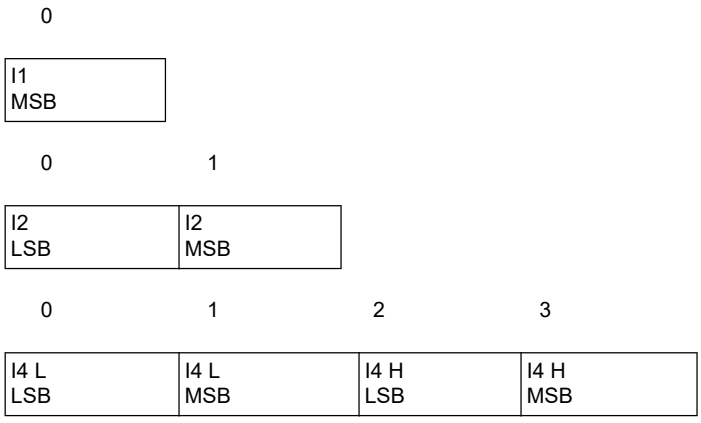
15.1.1 Unsigned Integer

Unsigned integers are stored without changes as 1-byte, 2-byte, or 4-byte values in Little Endian byte order.



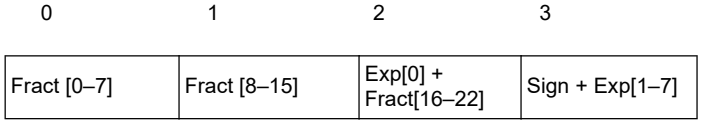
15.1.2 Integer

Integer values are stored in 2's-complement format in 1-byte, 2-byte, or 4-byte values in Little Endian byte order.



15.1.3 Floating Point

Floating point values are stored using the IEEE754 Single Precision 4-byte format in Little Endian byte order.



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Where:

Exp: 8-bit exponent stored with an offset bias of 127. The values 00 and FF have special meaning.

Fract: 23-bit fraction. If the exponent is > 0, then the mantissa is 1.fract. If the exponent is zero, then the mantissa is 0.fract.

The floating point value depends on the special cases of the exponent:

- If the exponent is FF and the fraction is zero, this represents +/– infinity.
- If the exponent is FF and the fraction is non-zero this represents "not a number" (NaN).
- If the exponent is 00 then the value is a subnormal number represented by $(-1)^{sign} \times 2^{-126} \times 0.fraction$.
- Otherwise, the value is a normalized number represented by $(-1)^{sign} \times 2^{(exponent - 127)} \times 1.fraction$.

15.1.4 Hex

Bit register definitions are stored in unsigned integer format.

15.1.5 String

String values are stored with length byte first, followed by a number of data bytes defined with the length byte.

0	1	. . .	N
Length	Data0	. . .	DataN

15.2 Calibration

15.2.1 Voltage

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Calibration	Voltage	Cell Gain	I2	–32768	32767	12101(1)	—	VC[n]–VC[n–1] gain
Calibration	Voltage	PACK Gain	U2	0	65535	49669(1)	—	PACK–VSS gain
Calibration	Voltage	VC2 (BAT) Gain	U2	0	65535	48936(1)	—	VC2 (BAT)–VSS gain

(1) Setting this value to 0 causes the gauge to use the internal factory calibration default.

15.2.2 Current

Class	Subclass	Name	Type	Min	Max	Default	Description
Calibration	Current	CC Gain	F4	1.00E–001	4.00E+000	3.58422	Coulomb Counter Gain
Calibration	Current	Capacity Gain	F4	2.98E+004	1.19E+006	106903.5	Capacity Gain

15.2.3 Current Offset

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Calibration	Current Offset	CC Offset	I2	–32768	32767	0	—	Coulomb Counter Offset
Calibration	Current Offset	Coulomb Counter Offset Samples	U2	0	65535	64	—	Coulomb Counter Offset Samples used for averaging
Calibration	Current Offset	Board Offset	I2	–32768	32767	0	—	PCB board offset
Calibration	Current Offset	CC Auto Config	H1	00	07	03	hex	Bit 0: AUTO_CAL_EN: Auto ***CC Offset*** calibration enable 0 = Disabled Auto CC calibration offset 1 = Firmware will perform auto CC calibration on entry into SLEEP mode. A min auto CC calibration interval is set to 10 hrs to prevent false wear out. The result is saved to ***CC Auto Offset*** . Bit 1: AUTO_NESTON: NEST Circuit ON 0 = HW NEST circuit is always on. Individual cell current measurement may have an error relative to *Current()*, but the *Current()* accuracy is not impacted. 1 = When ***[OFFSET_TAKEN]*** = 1, FW automatically controls the HW NEST circuit for best current and cell current measurements. Bit 2: OFFSET_TAKEN: ***CC Auto Offset*** is taken. 0 = ***CC Auto Offset*** has not been measured. 1 = ***CC Auto Offset*** has been measured. Bit 3 to Bit 7: Reserved

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Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Calibration	Current Offset	CC Auto Offset	I2	–10000	10000	0		***CC Offset*** collected via ***CC Auto Calibration*** . Used for cell current measurement and is different than ***CC Offset*** .

15.2.4 Current Deadband

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Calibration	Current Deadband	Deadband	U1	0	255	3	mA	Pack-based deadband to report 0 mA
Calibration	Current Deadband	Coulomb Counter Deadband	U1	0	255	9	116 nV	Coulomb counter deadband to report 0 charge (This setting should not be modified.)

15.2.5 Temperature

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Calibration	Temperature	Internal Temp Offset	I1	–128	127	0	0.1°C	Internal temperature sensor reading offset
Calibration	Temperature	External 1 Temp Offset	I1	–128	127	0	0.1°C	TS1 temperature sensor reading offset

15.2.6 Internal Temp Model

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Calibration	Internal Temp Model	Int Gain	I2	–32768	32768	–12143	—	Internal temperature gain
Calibration	Internal Temp Model	Int Base Offset	I2	–32768	32768	6232	—	Internal temperature base offset
Calibration	Internal Temp Model	Int Minimum AD	I2	–32768	32768	0	—	Minimum AD count used for calculation
Calibration	Internal Temp Model	Int Maximum Temp	I2	–32768	32768	6232	0.1 °K	Maximum Temperature boundary

15.2.7 Cell Temp Model

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Calibration	Cell Temp Model	Coefficient a1	I2	–32768	32768	–11130	—	Cell Temperature calculation polynomial a1
Calibration	Cell Temp Model	Coefficient a2	I2	–32768	32768	19142	—	Cell Temperature calculation polynomial a2
Calibration	Cell Temp Model	Coefficient a3	I2	–32768	32768	–19262	—	Cell Temperature calculation polynomial a3
Calibration	Cell Temp Model	Coefficient a4	I2	–32768	32768	28203	—	Cell Temperature calculation polynomial a4
Calibration	Cell Temp Model	Coefficient a5	I2	–32768	32768	892	—	Cell Temperature calculation polynomial a5
Calibration	Cell Temp Model	Coefficient b1	I2	–32768	32768	328	—	Cell Temperature calculation polynomial b1
Calibration	Cell Temp Model	Coefficient b2	I2	–32768	32768	–605	—	Cell Temperature calculation polynomial b2
Calibration	Cell Temp Model	Coefficient b3	I2	–32768	32768	–2443	—	Cell Temperature calculation polynomial b3
Calibration	Cell Temp Model	Coefficient b4	I2	–32768	32768	4969	—	Cell Temperature calculation polynomial b4
Calibration	Cell Temp Model	Rc0	I2	–32768	32768	11703	Ω	Resistance at 25°C
Calibration	Cell Temp Model	Adc0	I2	–32768	32768	11703	—	ADC reading at 25°C
Calibration	Cell Temp Model	Rpad	I2	–32768	32768	0[(1)]	Ω	Pad Resistance (0 to use factory calibration)
Calibration	Cell Temp Model	Rint	I2	–32768	32768	0[(1)]	Ω	Pullup resistor resistance (0 to use factory calibration)

(1) Setting this value to 0 causes the gauge to use the internal factory calibration default.

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15.3 Settings

15.3.1 Configuration

Class	Subclass	Name	Type	Min	Max	Default	Description
Settings	Configuration	FET Options	H1	0x00	0xFF	0x20	Bit 0: Reserved Bit 1: Reserved Bit 2: OTFET—FET action in OVERTEMPERATURE mode 0 = No FET action for overtemperature condition (default) 1 = CHG and DSG FETs will be turned off for overtemperature conditions. Bit 3: CHGSU—FET action in CHARGE SUSPEND mode 0 = FET active (default) 1 = Charging or Precharging disabled, FET off Bit 4: CHGIN—FET action in CHARGE INHIBIT mode 0 = FET active (default) 1 = Charging or Precharging disabled, FET off Bit 5: CHGFET—FET action on valid charge termination 0 = FET active (default) 1 = Charging or Precharging disabled, FET off Bit 6: SLEEPCHG—CHG FET enabled during sleep 0 = CHG FET off during sleep (default) 1 = CHG FET remains on during sleep Bit 7: PMPDRV—FET action in SLEEP mode (Refer to Appendix A .) 0 = AFE gate drive keep at 9.4 V in SLEEP mode (default) 1 = AFE gate drive automatically changes to 5.75 V in SLEEP mode.
Settings	Configuration	Gauging Configuration	H1	0x00	0x0F	0x04	Bit 0: RSOCL—*RelativeStateOfCharge()* and *RemainingCapacity()* behavior at end of charge 0 = Actual value shown (default) 1 = Held at 99% until valid charge termination. On entering valid charge termination update to 100% Bit 1: RSOC_HOLD—Prevent RSOC from increasing during discharge 0 = RSOC not limited 1 = RSOC not allowed to increase during discharge Bit 2: LOCK0—Keep *RemainingCapacity()* and *RelativeStateOfCharge()* from jumping back during relaxation after 0 was reached during discharge. 0 = Disabled (default) 1 = Enabled Bit 7:3: Reserved
Settings	Configuration	I2C Configuration	H1	0x00	0xFF	0x01	Bit 0: BCAST—Enable charging broadcast from device to smart charger 0 = Disabled 1 = Enabled (default) Bit 1: Reserved Bit 2: Reserved Bit 3: XL—Enable 400-kHz COM mode 0 = Normal bus speed (default) 1 = 400-kHz bus speed (slave mode) Bit 5:4: Reserved Bit 6: TO_STRETCH_EN—Enable bus timeouts (15-ms clock high and 25ms clock low) 0 = Disabled (default) 1 = Enabled Bit 7: FLASH_BUSY_WAIT—Enable clock stretching during a flash program or erase operation 0 = Disabled (default) 1 = Enabled
Settings	Configuration	Power Configuration	H1	0x00	0x01	0x00	Bit 0: AUTO_SHIP_EN—Automatically shuts down for shipment 0 = Disable auto shutdown feature (default) 1 = Enable auto shutdown after device is in SLEEP mode without communication for a set period of time. Bit 3: 1: Reserved Bit 4: SLP_ACCUM—Enable sleep charge accumulation 1 = Enable sleep charge accumulation 0 = Disable sleep charge accumulation (default) Bit 5: SLEEPWKCHG—Enable sleep wake charge feature 1 = Enable sleep wake charge feature 0 = Disable sleep wake charge feature (default) Bit 7: 6: Reserved

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Class	Subclass	Name	Type	Min	Max	Default	Description
Settings	Configuration	SOC Flag Config A	H2	0x0000	0xFFFF	0x0C8C	Bit 0: TDSETV—Enable TD flag set by cell voltage threshold 0 = Disabled (default) 1 = Enabled Bit 1: TDCLEARV—Enable TD flag clear by cell voltage threshold 0 = Disabled (default) 1 = Enabled Bit 2: TDSETRSOC—Enable TD flag set by RSOC threshold 0 = Disabled 1 = Enabled (default) Bit 3: TDCLEARRSOC—Enable TD flag clear by RSOC threshold 0 = Disabled 1 = Enabled (default) Bit 4: TCSETV—Enable TC flag set by cell voltage threshold 0 = Disabled (default) 1 = Enabled Bit 5: TCCLEARV—Enable TC flag clear by cell voltage threshold 0 = Disabled (default) 1 = Enabled Bit 6: TCSETRSOC—Enable TC flag set by RSOC threshold 0 = Disabled (default) 1 = Enabled Bit 7: TCCLEARRSOC—Enable TC flag clear by RSOC threshold 0 = Disabled 1 = Enabled (default) Bit 8: Reserved Bit 9: Reserved Bit 10: FCSETVCT—Enable FC flag set by primary charge termination 0 = Disabled 1 = Enabled (default) Bit 11: TCSETVCT—Enable TC flag set by primary charge termination 0 = Disabled 1 = Enabled (default) Bit 15: 12: Reserved
Settings	Configuration	SOC Flag Config B	H1	0x0000	0x00FF	0x008C	Bit 0: FDSETV—Enable FD flag set by cell voltage threshold 0 = Disabled (default) 1 = Enabled Bit 1: FDCLEARV—Enable FD flag clear by cell voltage threshold 0 = Disabled (default) 1 = Enabled Bit 2: FDSETRSOC—Enable FD flag set by RSOC threshold 0 = Disabled 1 = Enabled (default) Bit 3: FDCLEARRSOC—Enable FD flag clear by RSOC threshold 0 = Disabled 1 = Enabled (default) Bit 4: FCSETV—Enable FC flag set by cell voltage threshold 0 = Disabled (default) 1 = Enabled Bit 5: FCCLEARV—Enable FC flag clear by cell voltage threshold 0 = Disabled (default) 1 = Enabled Bit 6: FCSETRSOC—Enable FC flag set by RSOC threshold 0 = Disabled (default) 1 = Enabled Bit 7: FCCLEARRSOC—Enable FC flag clear by RSOC threshold 0 = Disabled 1 = Enabled (default)
Settings	Configuration	Charging Configuration	H1	0x00	0x3F	0x00	Bit 0: CRATE—ChargeCurrent rate 0 = No adjustment to *ChargingCurrent()* (default) 1 = *ChargingCurrent()* adjusted based on *FullChargeCapacity()*/ *DesignCapacity()* Bit 7:1: Reserved
Settings	Configuration	Temperature Enable	H1	0x00	0x03	0x03	Bit 0: internal TS— Enable Internal TS 0 = Disable internal TS (default) 1 = Enable internal TS Bit 1: TS1—Enable TS1 0 = Disable TS1 1 = Enable TS1 (default) Bit 7:2: Reserved
Settings	Configuration	DA Configuration	H1	0x00	0xFF	0x11	Bit 0: CC0—Cell Count 0 = 1 cell 1 = 2 cell Bit 1: Reserved Bit 2: Reserved Bit 3: IN_SYSTEM_SLEEP—In-system SLEEP mode 0 = Disable (default) 1 = Enable Bit 4: SLEEP—SLEEP Mode 0 = Disable SLEEP mode 1 = Enable SLEEP mode (default) Bit 5: Reserved Bit 6: CTEMP—Cell Temperature protection source 0 = MAX (default) 1 = Average Bit 7: Reserved

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Class	Subclass	Name	Type	Min	Max	Default	Description
Settings	Configuration	IT Gauging Configuration	H2	0x0000	0xFFFF	0xD4DE	Bit 0: CCT—Cycle count threshold 0 = Use CC % of *DesignCapacity()* (default) 1 = Use CC % of *FullChargeCapacity()* Bit 1: CSYNC—Sync *RemainingCapacity()* with *FullChargeCapacity()* at valid charge termination 0 = Not synchronized 1 = Synchronized (default) Bit 2: RFACTSTEP—Allow Ra update to limit before disqualifying further updates 0 = If (new Ra)/(old Ra) > 3, Ra update is not completed, and Ra updates are disabled. 1 = If (new Ra)/(old Ra) > 3, one Ra update is completed limited to factor of 3, and further Ra updates are disabled. Bit 3: OCVFR—Open Circuit Voltage Flat Region 0 = Disabled 1 = Enabled (default) Bit 4: Reserved Bit 5: Reserved Bit 6: RSOC_CONV—See Section 8.6. Bit 7: FAST_QMax_LRN—See Section 8.6. Bit 8: FAST_Qmax_FLD—See Section 8.6. Bit 9: CELL_TERM—See Section 8.6. Bit 10: FF_NEAR_EDV—See Section 8.6. Bit 11: RELAX_JUMP_OK—See Section 8.6 Bit 12: SMOOTH—See Section 8.6. Bit 13: Reserved Bit 14: Reserved Bit 15: SYNC_AT_OCV—See Section 8.6.
Settings	Configuration	Balancing Configuration	H1	0x00	0xFF	0x01	Bit 0: CB—Cell balancing 0 = Cell balancing disabled 1 = Cell balancing enabled (default) Bit 1: Reserved Bit 2: CBR—Cell balancing at rest 0 = Cell balancing at rest is disabled (default). 1 = Cell balancing at rest is enabled. Bit 7:3: Reserved

15.3.2 Charger

Note

The BROADCAST mode feature is available in the latest firmware version in the BQ28Z620 product folder on TI.com: It is not included in the shipped version.

The following data flash parameters enable the configuration of the BROADCAST mode feature. This feature is disabled by default because not all host MCU/CPUs are compatible with multi-master I^2C operation, and collisions could hang the host's comm engine.

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Settings	Charger	Device Address	H1	0	FF	D4	hex	Sets the address of slave device for BROADCAST mode
Settings	Charger	Voltage Register	H1	0	FF	0C	hex	BROADCAST mode: Sets the 2-byte address and data information to transmit to the slave device for charger output voltage setting
Settings	Charger	Current Register	H1	0	FF	0A	hex	BROADCAST mode: Sets the 2-byte address and data information to transmit to the slave device for charger output current setting
Settings	Charger	Broadcast Pacing	U1	0	255	15	s	BROADCAST mode: Period for broadcast

15.3.3 Protection

Class	Subclass	Name	Type	Min	Max	Default	Description
Settings	Protection	Protection Configuration	H1	0x00	0xFF	0x00	Bit 0: Reserved Bit 1: CUV_RECOV_CHG—Require charge to recover *SafetyStatus() [CUV]* 0 = Disabled 1 = Enabled (default) Bit 7:2: Reserved

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Class	Subclass	Name	Type	Min	Max	Default	Description
Settings	Protection	Enabled Protections A	H1	0x00	0xFF	0x57	Bit 0: CUV—Cell Undervoltage 0 = Disabled 1 = Enabled (default) Bit 1: COV—Cell Overvoltage 0 = Disabled 1 = Enabled (default) Bit 2: OCC—Overcurrent in Charge 0 = Disabled 1 = Enabled (default) Bit 3: Reserved Bit 4: OCD—Overcurrent in Discharge 0 = Disabled 1 = Enabled (default) Bit 5: Reserved Bit 6: Reserved (will be 1) Bit 7: Reserved
Settings	Protection	Enabled Protections B	H1	0x00	0xFF	0x35	Bit 0: Reserved (will be 1) Bit 1: Reserved Bit 2: Reserved (will be 1) Bit 3: Reserved Bit 4: OTC—Overtemperature in charge 0 = Disabled 1 = Enabled (default) Bit 5: OTD—Overtemperature in discharge 0 = Disabled 1 = Enabled (default) Bit 7:6: Reserved
Settings	Protection	Enabled Protections C	H1	0x00	0xFF	0x3C	Bit 0: Reserved Bit 1: Reserved Bit 2: PTO—Precharging timeout 0 = Disabled 1 = Enabled (default) Bit 3: Reserved Bit 4: CTO—Charging timeout 0 = Disabled 1 = Enabled (default) Bit 7:5: Reserved
Settings	Protection	Enabled Protections D	H1	0x00	0xFF	0x0C	Bit 0: Reserved Bit 1: Reserved Bit 2: UTC—Undertemperature while charging 0 = Disabled 1 = Enabled (default) Bit 3: UTD—Undertemperature while not charging 0 = Disabled 1 = Enabled (default) Bit 7:4: Reserved

15.3.4 Permanent Failure

Class	Subclass	Name	Type	Min	Max	Default	Description
Settings	Permanent Failure	Enabled PF A	H1	0x00	0xFF	0x00	Bit 0: SUV—Safety Cell Undervoltage 0 = Disabled (default) 1 = Enabled Bit 1: SOV—Safety Cell Overvoltage 0 = Disabled (default) 1 = Enabled Bit 7: 2: Reserved
Settings	Permanent Failure	Enabled PF B	H1	0x00	0xFF	0x00	Bit 2: 0: Reserved Bit 3: VIMR—Voltage Imbalance At Rest 0 = Disabled (default) 1 = Enabled Bit 4: VIMA—Voltage Imbalance At Active 0 = Disabled (default) 1 = Enabled Bit 7: 5: Reserved
Settings	Permanent Failure	Enabled PF C	H1	0x00	0xFF	0x00	Bit 0: CFETF—Charge FET Failure 0 = Disabled (default) 1 = Enabled Bit 1: DFETF—Discharge FET Failure 0 = Disabled (default) 1 = Enabled Bit 7: 2: Reserved

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15.3.5 AFE

Class	Subclass	Name	Type	Min	Max	Default	Description
Configuration	AFE	AFE Protection Control	H1	0x00	0xFF	0x00	Bit 0: RSNS—AOLD, ASCC, ASCD1, ASCD2 Thresholds 0 = 0.5 × AFE Protection Thresholds (default) 1 = Normal AFE Protection Thresholds Bit 1: SCDDx2—Double SCD Delay Times 0 = Normal SCD delay times (default) 1 = 2 × SCD delay times Bits 2–3: Reserved Bit 4–7: RSTRIM—"Unsupport" function. Should leave the default setting 0x7. Changing this setting may cause an error to the AFE current protection accuracy.
Configuration	AFE	ZVCHG Exit Threshold	I2	0	8000	2200	*Voltage()* threshold in mV when the gauge will exit ZVCHG mode when CFET is used for precharging.

15.3.6 Manufacturing

Class	Subclass	Name	Type	Min	Max	Default	Description
Settings	Manufacturing	Manufacturing Status	H2	0x0000	0xFFFF	0x0000	Bit 0: Reserved Bit 1: Reserved Bit 2: Reserved Bit 3: GAUGE_EN—Gauging 0 = Disabled (default) 1 = Enabled Bit 4: FET_EN—FET action 0 = Disabled (default) 1 = Enabled Bit 5: LF_EN—Lifetime data collection 0 = Disabled (default) 1 = Enabled Bit 6: PF_EN—Permanent failure 0 = Disabled (default) 1 = Enabled Bit 15:7: Reserved

15.4 Advanced Charging Algorithms

15.4.1 Temperature Ranges

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	Temperature Ranges	T1	I1	–128	127	0	°C	T1 low temperature range lower limit
Advanced Charging Algorithms	Temperature Ranges	T2	I1	–128	127	12	°C	T2 low temperature range to standard temperature range
Advanced Charging Algorithms	Temperature Ranges	T5	I1	–128	127	20	°C	T5 recommended temperature range lower limit
Advanced Charging Algorithms	Temperature Ranges	T6	I1	–128	127	25	°C	T6 recommended temperature range upper limit
Advanced Charging Algorithms	Temperature Ranges	T3	I1	–128	127	30	°C	T3 standard temperature range to high temperature range
Advanced Charging Algorithms	Temperature Ranges	T4	I1	–128	127	55	°C	T4 high temperature range upper limit
Advanced Charging Algorithms	Temperature Ranges	Hysteresis	I1	–128	127	1	°C	Temperature Hysteresis, applied when temperature is decreasing.

15.4.2 Low Temp Charging

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	Low Temp Charging	Voltage	I2	0	32767	4000	mV	Low temperature range *ChargingVoltage()*
Advanced Charging Algorithms	Low Temp Charging	Current Low	I2	0	32767	132	mA	Low temperature range low voltage range *ChargingCurrent()*

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Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	Low Temp Charging	Current Med	I2	0	32767	352	mA	Low temperature range medium voltage range *ChargingCurrent()*
Advanced Charging Algorithms	Low Temp Charging	Current High	I2	0	32767	264	mA	Low temperature range high voltage range *ChargingCurrent()*

15.4.3 Standard Temp Charging

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	Standard Temp Charging	Voltage	I2	0	32767	4200	mV	Standard temperature range *ChargingVoltage()*
Advanced Charging Algorithms	Standard Temp Charging	Current Low	I2	0	32767	1980	mA	Standard temperature range low voltage range *ChargingCurrent()*
Advanced Charging Algorithms	Standard Temp Charging	Current Med	I2	0	32767	4004	mA	Standard temperature range medium voltage range *ChargingCurrent()*
Advanced Charging Algorithms	Standard Temp Charging	Current High	I2	0	32767	2992	mA	Standard temperature range high voltage range *ChargingCurrent()*

15.4.4 High Temp Charging

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	High Temp Charging	Voltage	I2	0	32767	4000	mV	High temperature range *ChargingVoltage()*
Advanced Charging Algorithms	High Temp Charging	Current Low	I2	0	32767	1012	mA	High temperature range low voltage range *ChargingCurrent()*
Advanced Charging Algorithms	High Temp Charging	Current Med	I2	0	32767	1980	mA	High temperature range medium voltage range *ChargingCurrent()*
Advanced Charging Algorithms	High Temp Charging	Current High	I2	0	32767	1496	mA	High temperature range high voltage range *ChargingCurrent()*

15.4.5 Rec Temp Charging

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	Rec Temp Charging	Voltage	I2	0	32767	4100	mV	Recommended temperature range *ChargingVoltage()*
Advanced Charging Algorithms	Rec Temp Charging	Current Low	I2	0	32767	2508	mA	Recommended temperature range low voltage range *ChargingCurrent()*
Advanced Charging Algorithms	Rec Temp Charging	Current Med	I2	0	32767	4488	mA	Recommended temperature range medium voltage range *ChargingCurrent()*
Advanced Charging Algorithms	Rec Temp Charging	Current High	I2	0	32767	3520	mA	Recommended temperature range high voltage range *ChargingCurrent()*

15.4.6 Pre-Charging

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	PCHG	Current	I2	0	32767	88	mA	Precharge *ChargingCurrent()*

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15.4.7 Maintenance Charging

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	MCHG	Current	I2	0	32767	44	mA	Maintenance *ChargingCurrent()*

15.4.8 Voltage Range

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	Voltage Range	Threshold for entering precharge state	I2	0	32767	2500	mV	Minimum Cell voltage to enter PRECHARGE mode
Advanced Charging Algorithms	Voltage Range	Charging Voltage Low	I2	0	32767	2900	mV	Precharge Voltage range to Charging Voltage Low range
Advanced Charging Algorithms	Voltage Range	Charging Voltage Med	I2	0	32767	3600	mV	Charging Voltage Low range to Charging Voltage Med range
Advanced Charging Algorithms	Voltage Range	Charging Voltage High	I2	0	32767	4000	mV	Charging Voltage Med to Charging Voltage High range
Advanced Charging Algorithms	Voltage Range	Charging Voltage Hysteresis	U1	0	255	0	mV	Charging Voltage Hysteresis applied when voltage is decreasing

15.4.9 Termination Config

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	Termination Config	Charge Term Taper Current	I2	0	32767	250	mA	Valid Charge Termination taper current qualifier threshold
Advanced Charging Algorithms	Termination Config	Charge Term Voltage	I2	0	32767	75	mV	Valid Charge Termination delta voltage qualifier, max cell based

15.4.10 Cell Balancing Config

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Advanced Charging Algorithms	Cell Balancing Config	Balance Time per mAh cell 1	U2	0	65535	367	s/mAh	Required balance time per mAh. For information on how to calculate balancing time, see Section 9.1.
Advanced Charging Algorithms	Cell Balancing Config	Balance Time per mAh cell 2	U2	0	65535	514	s/mAh	Required balance time per mAh. For information on how to calculate balancing time, see Section 9.1.

15.5 Power

15.5.1 Power

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Power	Power	Valid Update Voltage	I2	0	32767	2800	mV	Min stack voltage threshold for Flash update

15.5.2 Shutdown

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Power	Shutdown	Shutdown Voltage	I2	0	32767	2300	mV	Cell based shutdown voltage trip threshold
Power	Shutdown	Shutdown Time	U2	0	255	10	s	Cell based shutdown voltage trip delay
Power	Shutdown	Charger Present Threshold	I2	0	32767	3000	mV	Pack terminal charger present detect threshold

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15.5.3 Sleep

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Power	Sleep	Sleep Current	I2	0	32767	10	mA	|*Current()*| threshold to enter SLEEP mode
Power	Sleep	Voltage Time	U1	0	255	5	s	Voltage sampling period in SLEEP mode
Power	Sleep	Current Time	U1	0	255	20	s	Current sampling period in SLEEP mode
Power	Sleep	Wake Comparator	H1	0x00	0xFF	0x00	—	Wake Comparator Configuration Setting Bits 7–4: Reserved Bits 3–2: Wk1, WK0—Wake Comparator Threshold 0,0 = ±0.625 mV 0,1 = ±1.25 mV 1,0 = ±2.5 mV 1,1 = ±5 mV Bits 1–0: Reserved

15.5.4 Ship

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Power	Ship	FET OFF time	U1	0	127	10	s	Delay time to turn OFF FETs prior to entering SHUTDOWN mode. This setting should not be longer than ***the Ship Delay*** setting.
Power	Ship	Delay	U1	0	254	20	s	Delay time to enter SHUTDOWN mode after FETs are turned OFF.
Power	Ship	Auto Ship Time	U2	0	65535	144	10 mins	The device will automatically enter SHUTDOWN mode after staying in SLEEP mode without communicating for this amount of time when ***Power Config[AUTO_SHIP_EN]*** = 1.

15.5.5 PMPDRV

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Power	PMPDRV	PMPDRV On VC2	I2	0	32767	3000	mV	When battery voltage on VC2 is too low, the AFE charge pump switches from 5.75 V to 9.4 V in SLEEP mode. (Refer to Appendix A.)
Power	PMPDRV	PMPDRV Hysteresis Voltage	I2	0	32767	200	mV	When battery voltage on VC2 is above ***PMPDRV On VC2*** + ***PMPDRV Hysteresis Voltage*** , the AFE charge pump switches back to 5.75 V in SLEEP mode.

15.6 Gas Gauging

15.6.1 Standby

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	Standby	Standby Current	I2	–32768	0	–10	mA	

15.6.2 Max Load

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	Max Load	Max Load Current	I2	–32768	32767	–500	mA	
Gas Gauging	Max Load	Max Load RSOC	U1	0	100	50	%	

15.6.3 Current Thresholds

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	Current Thresholds	Dsg Current Threshold	I2	–32768	32767	100	mA	DISCHARGE mode *Current()* threshold
Gas Gauging	Current Thresholds	Chg Current Threshold	I2	–32768	32767	50	mA	CHARGE mode *Current()* threshold
Gas Gauging	Current Thresholds	Quit Current	I2	0	32767	10	mA	|*Current()*| threshold to enter REST mode

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Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	Current Thresholds	Dsg Relax Time	U1	0	255	1	mA	Discharge to relax timeout
Gas Gauging	Current Thresholds	Chg Relax Time	U1	0	255	60	mA	Charge to relax timeout

15.6.4 Design

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	Design	Design Capacity mAh	I2	0	32767	4400	mAh	Design Capacity in mAh
Gas Gauging	Design	Design Capacity cWh	I2	0	32767	6336	cWh	Design Capacity in cWh
Gas Gauging	Design	Design Voltage	I2	0	32767	7200	mV	Design Voltage

15.6.5 Cycle

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	Cycle	Cycle Count Percentage	U1	0	100	90	%	Cycle Count Percentage

15.6.6 FD

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	FD	Set Voltage Threshold	I2	0	5000	3000	mV	*BatteryStatus()[FD]* cell voltage set threshold
Gas Gauging	FD	Clear Voltage Threshold	I2	0	5000	3100	mV	*BatteryStatus()[FD]* cell voltage clear threshold
Gas Gauging	FD	Set RSOC % Threshold	U1	0	100	0	%	*BatteryStatus()[FD]RelativeStateOfCharge()* set threshold
Gas Gauging	FD	Clear RSOC % Threshold	U1	0	100	5	%	*BatteryStatus()[FD]RelativeStateOfCharge()* clear threshold

15.6.7 FC

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	FC	Set Voltage Threshold	I2	0	5000	4200	mV	*BatteryStatus()[FC]* cell voltage set threshold
Gas Gauging	FC	Clear Voltage Threshold	I2	0	5000	4100	mV	*BatteryStatus()[FC]* cell voltage clear threshold
Gas Gauging	FC	Set RSOC % Threshold	U1	0	100	100	%	*BatteryStatus()[FC]RelativeStateOfCharge()* set threshold
Gas Gauging	FC	Clear RSOC % Threshold	U1	0	100	95	%	*BatteryStatus()[FC]RelativeStateOfCharge()* clear threshold

15.6.8 TDA

Per the *Smart Battery Data Specification v1.1*, TDA is only active while discharging.

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	TD	Set Voltage Threshold	I2	0	5000	3200	mV	*GaugingStatus[TD]* cell voltage set threshold
Gas Gauging	TD	Clear Voltage Threshold	I2	0	5000	3300	mV	*GaugingStatus[TD]* cell voltage clear threshold
Gas Gauging	TD	Set RSOC % Threshold	U1	0	100	6	%	*GaugingStatus[TD]RelativeStateOfCharge()* set threshold
Gas Gauging	TD	Clear RSOC % Threshold	U1	0	100	8	%	*GaugingStatus[TD]RelativeStateOfCharge()* clear threshold

15.6.9 TCA

Per the *Smart Battery Data Specification v1.1*, TCA is only active while charging.

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Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	TC	Set Voltage Threshold	I2	0	5000	4200	mV	*GaugingStatus[TC]* cell voltage set threshold
Gas Gauging	TC	Clear Voltage Threshold	I2	0	5000	4100	mV	*GaugingStatus[TC]* cell voltage clear threshold
Gas Gauging	TC	Set RSOC % Threshold	U1	0	100	100	%	*GaugingStatus[TC]RelativeStateOfCharge()* set threshold
Gas Gauging	TC	Clear RSOC % Threshold	U1	0	100	95	%	*GaugingStatus[TC]RelativeStateOfCharge()* clear threshold

15.6.10 State

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	State	QMax Cell 0	I2	0	32767	4400	mAh	QMax Cell 0
Gas Gauging	State	QMax Cell 1	I2	0	32767	4400	mAh	QMax Cell 1
Gas Gauging	State	QMax Pack	I2	0	32767	4400	mAh	QMax of the whole stack
Gas Gauging	State	QMax Cycle count	U2	0	65535	0		The *CycleCount()*when QMax updated
Gas Gauging	State	Update Status	H1	0x00	0xFF	0		Bit 1:0: Update1, Update0 Update Status 0,0 = Impedance Track gauging and lifetime updating is disabled. 0,1 = QMax updated 1,0 = QMax and Ra table have been updated Bit 2: Enable—Impedance Track gauging and lifetime updating is enabled. 0 = Disabled 1 = Enabled Bit 3: QMax update in the field 0 = Not updated 1 = Updated Bit 7:4: Reserved
Gas Gauging	State	Cell 0 Chg Voltage at EoC	I2	0	32767	4200	mV	Cell 0 voltage value at end of charge
Gas Gauging	State	Cell 1 Chg Voltage at EoC	I2	0	32767	4200	mV	Cell 1 voltage value at end of charge
Gas Gauging	State	Current at EoC	I2	–32768	32767	250	mA	Current at end of charge
Gas Gauging	State	Avg I Last Run	I2	–32768	32767	–2000	mA	Average current last discharge cycle
Gas Gauging	State	Avg P Last Run	I2	–32768	32767	–3022	10 mW	Average power last discharge cycle
Gas Gauging	State	Delta Voltage	I2	–32768	32767	0	mV	*Voltage()* delta between normal and short load spikes to optimize run time calculation
Gas Gauging	State	Temp k	I2	0	32767	100	0.1°C/ 256 cW	Initial thermal model temperature factor
Gas Gauging	State	Temp a	I2	0	32767	1000	—	Initial thermal model temperature
Gas Gauging	State	Max I Last Run	I2	–32768	32767	–2000	mA	Max current last discharge cycle
Gas Gauging	State	Max P Last Run	I2	–32768	32767	–3022	10 mW	Max power last discharge cycle
Gas Gauging	State	Cycle count	U2	0	65535	0	Cycle count	Value reported by *CycleCount()*. Updated by the gauge automatically when accumulated discharge exceeds the threshold set by ***Cycle Count Percentage***
Gas Gauging	State	SOH Temp k	I2	0	32767	100	0.1°C/ 256 cW	Initial thermal model temperature factor for SOH
Gas Gauging	State	SOH Temp a	I2	0	32767	1000	—	Initial thermal model temperature factor for SOH

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15.6.11 IT Config

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	IT Cfg	Pack Resistance	I2	0	32767	30	mΩ	
Gas Gauging	IT Cfg	System Resistance	I2	0	32767	0	mΩ	
Gas Gauging	IT Cfg	Ra Filter	U2	0	999	500	0.1%	Filter value used in Ra Updates, specifies what percentage or Ra update is from new value (100%—setting) vs. old value (setting). The recommended setting is 80% if the RSOC_CONV feature is enabled. Otherwise, the setting should be 50% as the default.
Gas Gauging	IT Cfg	Ra Max Delta	U1	0	255	15	% of Design Resistance	Maximum value of allowed Ra change
Gas Gauging	IT Cfg	Reference Grid	UI	0	15	4		***Reference Grid*** point used by Design Resistance. The default setting should be used if RSOC_CONV feature is enabled. Otherwise, grid point 11 should be used to ensure resistance updates fast enough at the grid where discharge termination occurs.
Gas Gauging	IT Cfg	Resistance Parameter Filter	U2	1	65534	65124	—	This is one of the filters used for resistance update. Reducing this filter setting can improve low temperature performance at high rates. The default setting is 41-s time constant. It is recommended to keep this filter within the range of 4 s (that is, DF setting = 61680) up to the default 41 s (that is, DF setting = 65142). Examining the Term Voltage Delta setting and Fast Scale Start SOC should be done prior to twisting this parameter when trying to improve the RSOC performance. The following is the formula to convert the DF setting into actual filter time constant: Filter time constant = [0.25/(1 – (DF_Value /65536))] – 0.25.
Gas Gauging	IT Cfg	Near EDV Ra Param Filter	U2	1	65535	59220	—	Ra filter used in the fast scaling region if [FF_NEAR_EDV] = 1. Default value should be used.
Gas Gauging	IT Cfg	QMax Delta	U1	3	100	5	%	Maximum allowed QMax change from its previous value. The QMax change will be capped by this setting if the delta from the previous QMax is larger than ***QMax Delta. QMax Delta*** is a percentage of Design Capacity.
Gas Gauging	IT Cfg	QMax Upper Bound	U1	100	255	130	%	Maximum QMax value over the lifetime of the pack. If the updated QMax value is larger than this setting, the updated QMax will be capped to ***QMax Upper Bound. QMax Upper Bound*** is a percentage of Design Capacity.
Gas Gauging	IT Cfg	Term Voltage	I2	0	32767	9000	mV	Min stack voltage to be used for capacity calculation
Gas Gauging	IT Cfg	Term Voltage Delta	I2	0	32767	300	mV	Controls when the RSOC_CONV feature becomes active. The recommended setting is 3.3 –Term Voltage/Number Cells. The default setting is 300 mV, which is assuming a typical 3-V termination voltage per cell. If a different termination voltage is used, this parameter should be adjusted accordingly.
Gas Gauging	IT Cfg	Max Simulation Iterations	U1	20	50	30		The maximum number of iterations allowed for simulation.
Gas Gauging	IT Cfg	Term Min Cell Voltage	I2	0	32767	2800	mV	Minimum cell termination voltage when used, if ***[CELL_TERM]*** = 1. This is intended to allow the IT algorithm to reach 0% before CUV is triggered; therefore, this value should be set at or above ***CUV:Threshold*** .
Gas Gauging	IT Cfg	Fast Scale Start SOC	U1	0	100	10	%	Control start of convergence when ***[RSOC_CONV]*** = 1 based on RSOC %. Raising this setting can improve RSOC drop at the end of discharge. However the RSOC % chosen for this setting must keep after the sharp drop of the discharge curve (the knee of the discharge curve).

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Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	IT Cfg	Load Select	U1	0	255	7		Defines Load compensation mode used by the gauging algorithm
Gas Gauging	IT Cfg	Load Mode	U1	0	255	0		Defines unit used by the gauging algorithm: 0 = Constant Current 1 = Constant Power
Gas Gauging	IT Cfg	Design Resistance						Averaged cell resistance at ***Reference Grid*** point. Automatically updated when Update Status is set to 0x6 by the gauge. To automatically update again, set Update Status to 0x4 or manually set when Update Status is set to 0x6.
Gas Gauging	IT Cfg	User-Rate-mA	I2	–32768	32768	0	mA	Discharge rate used for capacity calculation selected by Load Select
Gas Gauging	IT Cfg	User-Rate-mW	I2	–32768	32768	0	10 mW	Discharge rate used for capacity calculation selected by Load Select
Gas Gauging	IT Cfg	Reserve Cap-mAh	I2	0	9000	0	mAh	Capacity is reserved available when the gauging algorithm reports 0% *RelativeStateOfCharge()*.
Gas Gauging	IT Cfg	Reserve Cap-cWh	I2	0	32000	0	cWh	Capacity reserved is available when the gauging algorithm reports 0% *RelativeStateOfCharge()*.

15.6.12 Condition Flag

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	Condition Flag	Max Error Limit	U1	1	100	100	%	Max Error Limit Percentage

15.6.13 SOH

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	SOH	SOH Load Rate	U1	1	255	50	0.1 Hr rate	Current rate used in SOH simulation
Gas Gauging	SOH	Charge Voltage	I2	0	32767	4420	mV	Charge voltage specified in the battery cell data sheet for DOD estimation
Gas Gauging	SOH	Taper Current	I2	0	32767	250	mA	Taper current specified in the battery cell data sheet for DOD estimation

15.6.14 Max Error

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Gas Gauging	Max Error	Time Cycle Equivalent	U1	1	255	12	2h	After valid QMax update, each passed time period of Time Cycle Equivalent will increment of *MaxError()* by Cycle Delta. Time Cycle Equivalent increments Max Error by 0.05% for every Time Cycle Equivalent time period following the last QMax update.
Gas Gauging	Max Error	Cycle Delta	U1	0	255	5	0.01%	Each increment of *CycleCount()* after valid QMax update will increment of *MaxError()* by Cycle Delta. Setting this parameter to 0 disables the MaxError increment by time or cycle increment.

15.7 System Data

15.7.1 Manufacturer Data

Class	Subclass	Name	Type	Min	Max	Unit	Description
System Data	Manufacturer Data	Manufacturer Info A Length	U1	0	32	—	*ManufacturerInfo()* length
System Data	Manufacturer Data	Manufacturer Info Block A01-A32	U1				*ManufacturerInfo()* value

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15.7.2 Integrity

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
System Data	Integrity	Static DF Signature	H2	0	7FFF	0	hex	Status data flash signature. Use MAC *StaticDFSignature()* (with MSB set to 0) to initialize this value.
System Data	Integrity	Static Chem DF Signature	H2	0	7FFF	6C98	hex	Status Chemistry data signature. Use MAC *StaticChemDFSignature()* (with MSB set to 0) to initialize this value.
System Data	Integrity	All DF Signature	H2	0	7FFF	0	hex	Status data flash signature. Use MAC *AllDFSignature()* (with MSB set to 0) to initialize this value.

15.8 Configuration

15.8.1 Data

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Configuration	Data	Manufacturer Date	U2	0	65535	01/01/80		*ManufacturerDate()* value in the following format:Day + Month*32 + (Year–1980) * 512
Configuration	Data	Serial Number	H2	0x0000	0xFFFF	0x0001		*SerialNumber()* value
Configuration	Data	Manufacturer Name	S20+1	—	—	Texas Instruments	ASCII	*ManufacturerName()* value
Configuration	Data	Device Name	S20+1	—	—	BQ28Z620	ASCII	*DeviceName()* value
Configuration	Data	Device Chemistry	S4+1	—	—	LION	ASCII	*DeviceChemistry()* value

15.9 Lifetimes

15.9.1 Voltage

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Voltage	Max Voltage Cell 0	I2	0	32767	0	mV	Maximum reported cell voltage 0
Lifetimes	Voltage	Max Voltage Cell 1	I2	0	32767	0	mV	Maximum reported cell voltage 1

15.9.2 Current

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Current	Max Chg Current	I2	0	32768	0	200 mA	Maximum reported *Current()* in charge direction
Lifetimes	Current	Max Dsg Current	I2	–32768	0	0	200 mA	Maximum reported *Current()* in discharge direction

15.9.3 Temperature

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Temperature	Max Temp Cell	I1	–128	127	–128	°C	Maximum reported cell temperature
Lifetimes	Temperature	Min Temp Cell	I1	–128	127	127	°C	Minimum reported cell temperature

15.9.4 Safety Events

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Safety Events	No of COV Events	U1	0	255	0	8 events	Total number of *SafetyStatus()[COV]* events
Lifetimes	Safety Events	Last COV Event	U1	0	255	0	4 cycles	Last *SafetyStatus()[COV]* event in *CycleCount()* cycles
Lifetimes	Safety Events	No of CUV Events	U1	0	255	0	8 events	Total number of *SafetyStatus()[CUV]* events
Lifetimes	Safety Events	Last CUV Event	U1	0	255	0	4 cycles	Last *SafetyStatus()[CUV]* event in *CycleCount()* cycles
Lifetimes	Safety Events	No of OCC1 Events	U1	0	255	0	8 events	Total number of *SafetyStatus()[OCC1]* events
Lifetimes	Safety Events	Last OCC1 Event	U1	0	255	0	4 cycles	Last *SafetyStatus()[OCC1]* event in *CycleCount()* cycles

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Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Safety Events	No of ASCC Events	U1	0	255	0	8 events	Total number of *SafetyStatus()[SCC]* events
Lifetimes	Safety Events	Last ASCC Event	U1	0	255	0	4 cycles	Last *SafetyStatus()[SCC]* event in *CycleCount()* cycles

15.9.5 Power Events

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Power Events	No of Shutdowns	U1	0	255	0	events	Total number of shutdown events
Lifetimes	Power Events	No of Partial Resets	U1	0	255	0	events	Total number of partial reset events. (Refer to Section 6.4.1 for partial reset.)
Lifetimes	Power Events	No of Full Resets	U1	0	255	0	events	Total number of full reset events. (Refer to Section 6.4.1 for full reset.)
Lifetimes	Power Events	No of Watchdog Resets	U1	0	255	0	events	Total number of watchdog timer reset events

15.10 Protections

15.10.1 CUV—Cell Undervoltage

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	CUV	Threshold	I2	0	32767	2500	mV	Cell undervoltage trip threshold
Protections	CUV	Delay	U1	0	255	2	s	Cell undervoltage trip delay
Protections	CUV	Recovery	I2	0	32767	3000	mV	Cell undervoltage recovery threshold

15.10.2 COV—Cell Overvoltage

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	COV	Threshold Low Temp	I2	0	32767	4300	mV	Cell overvoltage low temperature range trip threshold
Protections	COV	Threshold Standard Temp	I2	0	32767	4300	mV	Cell overvoltage standard temperature range trip threshold
Protections	COV	Threshold High Temp	I2	0	32767	4300	mV	Cell overvoltage high temperature range trip threshold
Protections	COV	Threshold Rec Temp	I2	0	32767	4300	mV	Cell overvoltage recommended temperature range trip threshold
Protections	COV	Delay	U1	0	255	2	s	Cell overvoltage trip delay
Protections	COV	Recovery Low Temp	I2	0	32767	3900	mV	Cell overvoltage low temperature range recovery threshold
Protections	COV	Recovery Standard Temp	I2	0	32767	3900	mV	Cell overvoltage standard temperature recovery range threshold
Protections	COV	Recovery High Temp	I2	0	32767	3900	mV	Cell overvoltage high temperature range recovery threshold
Protections	COV	Recovery Rec Temp	I2	0	32767	3900	mV	Cell overvoltage recommended temperature range recovery threshold

15.10.3 OCC—Overcurrent In Charge

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	OCC	Threshold	I2	–32768	32767	6000	mA	Overcurrent in Charge trip threshold
Protections	OCC	Delay	U1	0	255	0	s	Overcurrent in Charge trip delay
Protections	OCC	Recovery Threshold	I2	–32768	32767	200	mA	Overcurrent in Charge recovery threshold
Protections	OCC	Recovery Delay	U1	0	255	5	s	Overcurrent in Charge recovery delay

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15.10.4 OCD—Overcurrent In Discharge

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	OCD	Threshold	I2	–32768	32767	–6000	mA	Overcurrent in Discharge trip threshold
Protections	OCD	Delay	U1	0	255	6	s	Overcurrent in Discharge trip delay
Protections	OCD	Recovery	I2	–32768	32767	50	mA	Overcurrent in Discharge recovery threshold
Protections	OCD	Recovery Delay	U1	0	255	5	s	Overcurrent in Discharge recovery delay

15.10.5 AOLD—AFE Over Load In Discharge

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	AOLD	Threshold	H1	0x00	0xFF	0xF4	hex	***AOLD: Threshold*** Setting, Bits 7–4: OLDD: AOLD delay time Setting, Bits 3–0: OLDV: AOLD threshold
Protections	AOLD	Recovery	U1	0	255	5	s	Overload recovery time

15.10.6 ASCC—AFE Short Circuit in Charge

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	ASCC	Threshold	H1	0x00	0xFF	0x77	hex	***ASCC: Threshold*** Setting, Bits 7–4: SCCD: ASCC delay time Setting, Bit 3: Reserved Setting, Bits 2–0: OLDV: AOLD threshold Setting
Protections	ASCC	Recovery	U1	0	255	5	s	Overload recovery time

15.10.7 ASCD—AFE Short Circuit in Discharge

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	ASCD	Threshold 1	H1	0x00	0xFF	0x77	hex	***ASCD: Threshold 1*** Setting, Bits 7–4: SCD1D: SCD1 delay time Setting, Bit 3: Reserved Setting, Bits 2–0: SCD1V: SCD1 threshold setting
Protections	ASCD	Threshold 2	H1	0x00	0xFF	0xE7	hex	***ASCD: Threshold 2*** Setting, Bits 7–4: SCD2D: SCD2 delay time Setting, Bit 3: Reserved Setting, Bits 2–0: SCD2V: SCD2 threshold setting
Protections	ASCD	Recovery	U1	0	255	5	s	Overload recovery time

15.10.8 OTC—Overtemperature in Charge

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	OTC	Threshold	I2	–400	1500	550	0.1°C	Overtemperature in Charge trip threshold
Protections	OTC	Delay	U1	0	255	2	s	Overtemperature in Charge Cell trip delay
Protections	OTC	Recovery	I2	–400	1500	500	0.1°C	Overtemperature in Charge Cell recovery threshold

15.10.9 OTD—Overtemperature in Discharge

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	OTD	Threshold	I2	–400	1500	600	0.1°C	Overtemperature in Discharge trip threshold
Protections	OTD	Delay	U1	0	255	2	s	Overtemperature in Discharge trip delay
Protections	OTD	Recovery	I2	–400	1500	550	0.1°C	Overtemperature in Discharge recovery threshold

15.10.10 UTC—Under Temperature in Charge

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	UTC	Threshold	I2	–400	1500	0	0.1°C	Under Temperature in Charge trip threshold
Protections	UTC	Delay	U1	0	255	2	s	Under Temperature in Charge Cell trip delay

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Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	UTC	Recovery	I2	–400	1500	50	0.1°C	Under Temperature in Charge Cell recovery threshold

15.10.11 UTD—Under Temperature in Discharge

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	UTD	Threshold	I2	–400	1500	0	0.1°C	Under Temperature in Discharge trip threshold
Protections	UTD	Delay	U1	0	255	2	s	Under Temperature in Discharge trip delay
Protections	UTD	Recovery	I2	–400	1500	50	0.1°C	Under Temperature in Discharge recovery threshold

15.10.12 PTO—PRECHARGE Mode Time Out

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	PTO	Charge Threshold	I2	–32768	32767	2000	mA	Precharge Timeout Current Threshold
Protections	PTO	Suspend Threshold	I2	–32768	32767	1800	mA	Precharge Timeout Suspend Threshold
Protections	PTO	Delay	U2	0	65535	1800	s	Precharge Timeout trip delay
Protections	PTO	Reset	I2	–32768	32767	2	mAh	Precharge Timeout Reset Threshold

15.10.13 CTO—FAST CHARGE Mode Time Out

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	CTO	Charge Threshold	I2	–32768	32767	2500	mA	Fast Charge Timeout Current Threshold
Protections	CTO	Suspend Threshold	I2	–32768	32767	2000	mA	Fast Charge Timeout Suspend Threshold
Protections	CTO	Delay	U2	0	65535	54000	s	Fast Charge Timeout trip delay
Protections	CTO	Reset	I2	–32768	32767	2	mAh	Fast Charge Timeout Reset Threshold

15.11 Permanent Fail

15.11.1 SUV—Safety Cell Undervoltage

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Permanent Fail	SUV	Threshold	I2	0	32767	1000	mV	Threshold below which the permanent fail alert is triggered
Permanent Fail	SUV	Delay	U1	0	255	5	s	Delay until the permanent fail trip is triggered

15.11.2 SOV—Safety Cell Overvoltage

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Permanent Fail	SOV	Threshold	I2	0	32767	4500	mV	Safety Cell Overvoltage trip threshold
Permanent Fail	SOV	Delay	U1	0	255	5	s	Safety Cell Overvoltage trip delay

15.11.3 VIMR—Voltage Imbalance at Rest

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Permanent Fail	VIMR	Check Voltage	I2	0	5000	3600	mV	Voltage Imbalance at Rest Check Voltage
Permanent Fail	VIMR	Check Current	I2	0	32767	10	mA	Voltage Imbalance at Rest Check Current
Permanent Fail	VIMR	Delta Threshold	I2	0	5000	200	mV	Voltage Imbalance at Rest Trip Threshold
Permanent Fail	VIMR	Delay	U1	0	255	2	s	Voltage Imbalance at Rest Check Trip Delay

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Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Permanent Fail	VIMR	Duration	U2	0	65535	100	s	Voltage Imbalance at Rest Check Duration

15.11.4 VIMA—Voltage Imbalance Active

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Permanent Fail	VIMA	Check Voltage	I2	0	5000	3600	mV	Voltage Imbalance active Check Voltage
Permanent Fail	VIMA	Check Current	I2	0	32767	10	mA	Voltage Imbalance active Check Current
Permanent Fail	VIMA	Delta Threshold	I2	0	5000	300	mV	Voltage Imbalance active trip threshold
Permanent Fail	VIMA	Delay	U1	0	255	2	s	Voltage Imbalance active Check trip delay

15.12 PF Status

The data in this class is saved at the time of the PF event.

15.12.1 Device Status Data

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Status Data	Safety Alert A	H1	0	FF	0	hex	Accumulated safety flags since PF event
PF Status	Device Status Data	Safety Status A	H1	0	FF	0	hex	Accumulated safety flags since PF event

Name: Safety Alert A or Safety Status A							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
RSVD	AOLD	RSVD	OCD	RSVD	OCC	COV	CUV

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Status Data	Safety Alert B	H1	0	FF	0	hex	Accumulated safety flags since PF event
PF Status	Device Status Data	Safety Status B	H1	0	FF	0	hex	Accumulated safety flags since PF event

Name: Safety Alert B or Safety Status B							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
RSVD	RSVD	OTD	OTC	RSVD	ASCD	RSVD	ASCC

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Status Data	Safety Alert C	H1	0	FF	0	hex	Accumulated safety flags since PF event
PF Status	Device Status Data	Safety Status C	H1	0	FF	0	hex	Accumulated safety flags since PF event

Name: Safety Alert C or Safety Status C							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
RSVD	RSVD	CTOS	RSVD	PTOS	RSVD	RSVD	RSVD

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Status Data	Safety Alert D	H1	0	FF	0	hex	Accumulated safety flags since PF event. All bits in this register are RSVD.
PF Status	Device Status Data	Safety Status D	H1	0	FF	0	hex	Accumulated safety flags since PF event

Name: Safety Status D							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
RSVD	RSVD	CTOS	RSVD	PTOS	DFW	RSVD	IFC

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Status Data	Operation Status A	H2	0	FFFF	0	hex	*OperationStatus()* data at the time of the PF event

Name: Operation Status A															
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
RSVD	RSVD	RSVD	RSVD	PCHG	CHG	DSG	RSVD	SLEEP	XCHG	XDSG	PF	SS	SDV	SEC1	SEC0

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Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Status Data	Operation Status B	H2	0	FFFF	0	hex	*OperationStatus()* data at the time of the PF event

Name: Operation Status B															
Bit 15	**Bit 14**	**Bit 13**	**Bit 12**	**Bit 11**	**Bit 10**	**Bit 9**	**Bit 8**	**Bit 7**	**Bit 6**	**Bit 5**	**Bit 4**	**Bit 3**	**Bit 2**	**Bit 1**	**Bit 0**
SLEEPM	XL	CAL_OFFSET	CAL	AUTO CALM	AUTH	LED	SDM	RSVD	RSVD	RSVD	CB	SLPCC	SLPAD	SMBL CAL	INIT

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Status Data	Temp Range	H1	0	FF	0	hex	Temperature range status at the time of the PF event. The temperature range information is returned to *ChargingStatus()*.

Name: Temp Range							
Bit 7	**Bit 6**	**Bit 5**	**Bit 4**	**Bit 3**	**Bit 2**	**Bit 1**	**Bit 0**
RSVD	OT	HT	STH	RT	SLT	LT	UT

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Status Data	Charging Status A	H1	0	FF	0	hex	The charging status at the time of the PF event. See section under *MACSubcmd()*, *Gauging Status()*.

Name: Charging Status A							
Bit 7	**Bit 6**	**Bit 5**	**Bit 4**	**Bit 3**	**Bit 2**	**Bit 1**	**Bit 0**
VCT	MCHG	SU	IN	HV	MV	LV	PV

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Status Data	Gauging Status	H1	0	FF	0	hex	The charging status at the time of the PF event. See section under *MACSubcmd()*, *Gauging Status()*.

Name: Gauging Status							
Bit 7	**Bit 6**	**Bit 5**	**Bit 4**	**Bit 3**	**Bit 2**	**Bit 1**	**Bit 0**
VCT	MCHG	SU	IN	HV	MV	LV	PV

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Status Data	IT Status	H2	0	FFFF	0	hex	The Impedance Track status at the time of the PF event. See section under *MACSubcmd()*, *Gauging Status()*.

Name: IT Status															
Bit 15	**Bit 14**	**Bit 13**	**Bit 12**	**Bit 11**	**Bit 10**	**Bit 9**	**Bit 8**	**Bit 7**	**Bit 6**	**Bit 5**	**Bit 4**	**Bit 3**	**Bit 2**	**Bit 1**	**Bit 0**
NSFM	RSVD	SLPQMax	QEN	VOK	RDIS	RSVD	REST	RSVD	RSVD	RSVD	OCVFR	LDMD	RX	QMax	VDQ

15.12.2 Device Voltage Data

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Voltage Data	Cell Voltage 0	I2	–32768	32767	0	mV	Cell 0 voltage
PF Status	Device Voltage Data	Cell Voltage 1	I2	–32768	32767	0	mV	Cell 1 voltage
PF Status	Device Voltage Data	Bat Direct Voltage	I2	–32768	32767	0	mV	Cell stack voltage
PF Status	Device Voltage Data	Pack Voltage	I2	–32768	32767	0	mV	Pack terminal voltage

15.12.3 Device Current Data

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Current Data	Current	I2	–32768	32767	0	mA	*Current()*

15.12.4 Device Temperature Data

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Temperature Data	Internal Temperature	I2	–32768	32767	0	0.1 K	Internal temperature sensor Temperature
PF Status	Device Temperature Data	External 1 Temperature	I2	–32768	32767	0	0.1 K	External TS1 Temperature

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15.12.5 Device Gauging Data

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Gauging Data	Cell 0 DOD0	I2	–32768	32767	0		Cell 0 depth of discharge
PF Status	Device Gauging Data	Cell 1 DOD0	I2	–32768	32767	0		Cell 1 depth of discharge
PF Status	Device Gauging Data	Passed Charge	I2	–32768	32767	0	mAh	Passed charge since last QMax update

15.12.6 AFE Registers

Class	Subclass	Name	Type	Min	Max	Default	Description
PF Status	AFE Regs	AFE Interrupt Status	H1	0x00	0xFF	0x00	AFE Interrupt Status Register Contents
PF Status	AFE Regs	AFE FET Status	H1	0x00	0xFF	0x00	AFE FET Status Register Contents
PF Status	AFE Regs	AFE Rxin	H1	0x00	0xFF	0x00	AFE Rxin Register Contents
PF Status	AFE Regs	AFE Interrupt Enable	H1	0x00	0xFF	0x00	AFE Interrupt Enable Register Contents
PF Status	AFE Regs	AFE FET Control	H1	0x00	0xFF	0x00	AFE FET Control Register Contents
PF Status	AFE Regs	AFE RXIEN	H1	0x00	0xFF	0x00	AFE RXIEN Register Contents
PF Status	AFE Regs	AFE RLOUT	H1	0x00	0xFF	0x00	AFE RLOUT Register Contents
PF Status	AFE Regs	AFE RHOUT	H1	0x00	0xFF	0x00	AFE RHOUT Register Contents
PF Status	AFE Regs	AFE RHINT	H1	0x00	0xFF	0x00	AFE RHINT Register Contents
PF Status	AFE Regs	AFE Cell Balance	H1	0x00	0xFF	0x00	AFE Cell Balance Register Contents
PF Status	AFE Regs	AFE AD/CC Control	H1	0x00	0xFF	0x00	AFE AD/CC Control Register Contents
PF Status	AFE Regs	AFE ADC Mux	H1	0x00	0xFF	0x00	AFE ADC Mux Register Contents
PF Status	AFE Regs	AFE State Control	H1	0x00	0xFF	0x00	AFE State Control Register Contents
PF Status	AFE Regs	AFE Wake Control	H1	0x00	0xFF	0x00	AFE Wake Control Register Contents
PF Status	AFE Regs	AFE Protection Control	H1	0x00	0xFF	0x00	AFE Protection Control Register Contents
PF Status	AFE Regs	AFE OCD	H1	0x00	0xFF	0x00	AFE OCD Register Contents
PF Status	AFE Regs	AFE SCC	H1	0x00	0xFF	0x00	AFE SCC Register Contents
PF Status	AFE Regs	AFE SCD1	H1	0x00	0xFF	0x00	AFE SCD1 Register Contents
PF Status	AFE Regs	AFE SCD2	H1	0x00	0xFF	0x00	AFE SCD2 Register Contents

15.13 RA Table

15.13.1 R_a0

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
RA Table	R_a0	Cell 0 R_A Flag	H2	0x0000	0xFFFF	0xFF55		High Byte: 0x00: Cell Impedance and QMax updated 0x05: RELAX mode and QMax update in progress 0x55: DISCHARGE mode and cell updated 0xFF: Cell impedance never updated Low-Byte: 0x00: Table not used and QMax updated 0x55: Table being used 0xFF: Table never used, no QMax or cell impedance update
RA Table	R_a0	Cell 0 R_A 0	I2	–32768	32768	38	2^{-10} Ω	Cell 0 resistance at grid point 0
RA Table	R_a0	Cell 0 R_A 1	I2	–32768	32768	41	2^{-10} Ω	Cell 0 resistance at grid point 1
RA Table	R_a0	Cell 0 R_A 2	I2	–32768	32768	43	2^{-10} Ω	Cell 0 resistance at grid point 2
RA Table	R_a0	Cell 0 R_A 3	I2	–32768	32768	44	2^{-10} Ω	Cell 0 resistance at grid point 3
RA Table	R_a0	Cell 0 R_A 4	I2	–32768	32768	42	2^{-10} Ω	Cell 0 resistance at grid point 4
RA Table	R_a0	Cell 0 R_A 5	I2	–32768	32768	42	2^{-10} Ω	Cell 0 resistance at grid point 5
RA Table	R_a0	Cell 0 R_A 6	I2	–32768	32768	45	2^{-10} Ω	Cell 0 resistance at grid point 6
RA Table	R_a0	Cell 0 R_A 7	I2	–32768	32768	48	2^{-10} Ω	Cell 0 resistance at grid point 7
RA Table	R_a0	Cell 0 R_A 8	I2	–32768	32768	49	2^{-10} Ω	Cell 0 resistance at grid point 8
RA Table	R_a0	Cell 0 R_A 9	I2	–32768	32768	52	2^{-10} Ω	Cell 0 resistance at grid point 9
RA Table	R_a0	Cell 0 R_A 10	I2	–32768	32768	56	2^{-10} Ω	Cell 0 resistance at grid point 10
RA Table	R_a0	Cell 0 R_A 11	I2	–32768	32768	64	2^{-10} Ω	Cell 0 resistance at grid point 11
RA Table	R_a0	Cell 0 R_A 12	I2	–32768	32768	74	2^{-10} Ω	Cell 0 resistance at grid point 12
RA Table	R_a0	Cell 0 R_A 13	I2	–32768	32768	128	2^{-10} Ω	Cell 0 resistance at grid point 13
RA Table	R_a0	Cell 0 R_A 14	I2	–32768	32768	378	2^{-10} Ω	Cell 0 resistance at grid point 14

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15.13.2 R_a1

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
RA Table	R_a1	Cell 1 R_A Flag	H2	0x0000	0xFFFF	0xFF55		High-Byte: 0x00: Cell Impedance and QMax updated 0x05: RELAX mode and QMax update in progress 0x55: DISCHARGE mode and cell updated 0xFF: cell impedance never updated Low-Byte: 0x00: Table not used and QMax updated 0x55: Table being used 0xFF: Table never used, no QMax or cell impedance update
RA Table	R_a1	Cell 1 R_A 0	I2	–32768	32768	38	2^{-10} Ω	Cell 1 resistance at grid point 0
RA Table	R_a1	Cell 1 R_A 1	I2	–32768	32768	41	2^{-10} Ω	Cell 1 resistance at grid point 1
RA Table	R_a1	Cell 1 R_A 2	I2	–32768	32768	43	2^{-10} Ω	Cell 1 resistance at grid point 2
RA Table	R_a1	Cell 1 R_A 3	I2	–32768	32768	44	2^{-10} Ω	Cell 1 resistance at grid point 3
RA Table	R_a1	Cell 1 R_A 4	I2	–32768	32768	42	2^{-10} Ω	Cell 1 resistance at grid point 4
RA Table	R_a1	Cell 1 R_A 5	I2	–32768	32768	42	2^{-10} Ω	Cell 1 resistance at grid point 5
RA Table	R_a1	Cell 1 R_A 6	I2	–32768	32768	45	2^{-10} Ω	Cell 1 resistance at grid point 6
RA Table	R_a1	Cell 1 R_A 7	I2	–32768	32768	48	2^{-10} Ω	Cell 1 resistance at grid point 7
RA Table	R_a1	Cell 1 R_A 8	I2	–32768	32768	49	2^{-10} Ω	Cell 1 resistance at grid point 8
RA Table	R_a1	Cell 1 R_A 9	I2	–32768	32768	52	2^{-10} Ω	Cell 1 resistance at grid point 9
RA Table	R_a1	Cell 1 R_A 10	I2	–32768	32768	56	2^{-10} Ω	Cell 1 resistance at grid point 10
RA Table	R_a1	Cell 1 R_A 11	I2	–32768	32768	64	2^{-10} Ω	Cell 1 resistance at grid point 11
RA Table	R_a1	Cell 1 R_A 12	I2	–32768	32768	74	2^{-10} Ω	Cell 1 resistance at grid point 12
RA Table	R_a1	Cell 1 R_A 13	I2	–32768	32768	128	2^{-10} Ω	Cell 1 resistance at grid point 13
RA Table	R_a1	Cell 1 R_A 14	I2	–32768	32768	378	2^{-10} Ω	Cell 1 resistance at grid point 14

15.13.3 R_a0x

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
RA Table	R_a0x	xCell 0 R_A Flag	H2	0x0000	0xFFFF	0xFFFF		High-Byte: 0x00: Cell Impedance and QMax updated 0x05: RELAX mode and QMax update in progress 0x55: DISCHARGE mode and cell updated 0xFF: Cell impedance never updated. Low-Byte: 0x00: Table is not used and QMax updated. 0x55: Table is being used. 0xFF: Table was never used and no QMax or cell impedance was updated.
RA Table	R_a0x	xCell 0 R_A 0	I2	–32768	32768	38	2^{-10} Ω	Cell 0 resistance at grid point 0
RA Table	R_a0x	xCell 0 R_A 1	I2	–32768	32768	41	2^{-10} Ω	Cell 0 resistance at grid point 1
RA Table	R_a0x	xCell 0 R_A 2	I2	–32768	32768	43	2^{-10} Ω	Cell 0 resistance at grid point 2
RA Table	R_a0x	xCell 0 R_A 3	I2	–32768	32768	44	2^{-10} Ω	Cell 0 resistance at grid point 3
RA Table	R_a0x	xCell 0 R_A 4	I2	–32768	32768	42	2^{-10} Ω	Cell 0 resistance at grid point 4
RA Table	R_a0x	xCell 0 R_A 5	I2	–32768	32768	42	2^{-10} Ω	Cell 0 resistance at grid point 5
RA Table	R_a0x	xCell 0 R_A 6	I2	–32768	32768	45	2^{-10} Ω	Cell 0 resistance at grid point 6
RA Table	R_a0x	xCell 0 R_A 7	I2	–32768	32768	48	2^{-10} Ω	Cell 0 resistance at grid point 7
RA Table	R_a0x	xCell 0 R_A 8	I2	–32768	32768	49	2^{-10} Ω	Cell 0 resistance at grid point 8
RA Table	R_a0x	xCell 0 R_A 9	I2	–32768	32768	52	2^{-10} Ω	Cell 0 resistance at grid point 9
RA Table	R_a0x	xCell 0 R_A 10	I2	–32768	32768	56	2^{-10} Ω	Cell 0 resistance at grid point 10
RA Table	R_a0x	xCell 0 R_A 11	I2	–32768	32768	64	2^{-10} Ω	Cell 0 resistance at grid point 11
RA Table	R_a0x	xCell 0 R_A 12	I2	–32768	32768	74	2^{-10} Ω	Cell 0 resistance at grid point 12
RA Table	R_a0x	xCell 0 R_A 13	I2	–32768	32768	128	2^{-10} Ω	Cell 0 resistance at grid point 13
RA Table	R_a0x	xCell 0 R_A 14	I2	–32768	32768	378	2^{-10} Ω	Cell 0 resistance at grid point 14

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15.13.4 R_a1x

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
RA Table	R_a1x	xCell 1 R_A Flag	H2	0x0000	0xFFFF	0xFFFF		High-Byte: 0x00: Cell Impedance and QMax updated 0x05: RELAX mode and QMax update in progress 0x55: DISCHARGE mode and Cell updated 0xFF: Cell impedance never updated Low-Byte: 0x00: Table is not used and QMax updated. 0x55: Table is being used. 0xFF: Table was never used and no QMax or cell impedance was updated.
RA Table	R_a1x	xCell 1 R_A 0	I2	–32768	32768	38	2^{-10} Ω	Cell 1 resistance at grid point 0
RA Table	R_a1x	xCell 1 R_A 1	I2	–32768	32768	41	2^{-10} Ω	Cell 1 resistance at grid point 1
RA Table	R_a1x	xCell 1 R_A 2	I2	–32768	32768	43	2^{-10} Ω	Cell 1 resistance at grid point 2
RA Table	R_a1x	xCell 1 R_A 3	I2	–32768	32768	44	2^{-10} Ω	Cell 1 resistance at grid point 3
RA Table	R_a1x	xCell 1 R_A 4	I2	–32768	32768	42	2^{-10} Ω	Cell 1 resistance at grid point 4
RA Table	R_a1x	xCell 1 R_A 5	I2	–32768	32768	42	2^{-10} Ω	Cell 1 resistance at grid point 5
RA Table	R_a1x	xCell 1 R_A 6	I2	–32768	32768	45	2^{-10} Ω	Cell 1 resistance at grid point 6
RA Table	R_a1x	xCell 1 R_A 7	I2	–32768	32768	48	2^{-10} Ω	Cell 1 resistance at grid point 7
RA Table	R_a1x	xCell 1 R_A 8	I2	–32768	32768	49	2^{-10} Ω	Cell 1 resistance at grid point 8
RA Table	R_a1x	xCell 1 R_A 9	I2	–32768	32768	52	2^{-10} Ω	Cell 1 resistance at grid point 9
RA Table	R_a1x	xCell 1 R_A 10	I2	–32768	32768	56	2^{-10} Ω	Cell 1 resistance at grid point 10
RA Table	R_a1x	xCell 1 R_A 11	I2	–32768	32768	64	2^{-10} Ω	Cell 1 resistance at grid point 11
RA Table	R_a1x	xCell 1 R_A 12	I2	–32768	32768	74	2^{-10} Ω	Cell 1 resistance at grid point 12
RA Table	R_a1x	xCell 1 R_A 13	I2	–32768	32768	128	2^{-10} Ω	Cell 1 resistance at grid point 13
RA Table	R_a1x	xCell 1 R_A 14	I2	–32768	32768	378	2^{-10} Ω	Cell 1 resistance at grid point 14

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16.1 Data Flash Table

Table 16-1. Data Flash Table

Class	Subclass	Address	Name	Type	Min Value	Max Value	Default	Units
Calibration	Voltage	0x4000	Cell Gain	I2	–32767	32767	12101	—
Calibration	Voltage	0x4002	Pack Gain	U2	0	65535	49669	—
Calibration	Voltage	0x4004	BAT Gain	U2	0	65535	48936	—
Calibration	Current	0x4006	CC Gain	F4	1.00E–01	4.00E+00	3.58422	—
Calibration	Current	0x400A	Capacity Gain	F4	2.98262E+04	1.193046E+06	1069035.256	—
Calibration	Current Offset	0x400E	CC Offset	I2	–32767	32767	0	—
Calibration	Current Offset	0x4010	Coulomb Counter Offset Samples	U2	0	65535	64	—
Calibration	Current Offset	0x4012	Board Offset	I2	–32768	32767	0	—
Calibration	Current Offset	0x40C0	CC Auto Config	H1	0x00	0x07	0x03	Hex
Calibration	Current Offset	0x40C1	CC Auto Offset	I2	–10000	10000	0	—
Calibration	Temperature	0x4014	Internal Temp Offset	I1	–128	127	0	0.1°C
Calibration	Temperature	0x4015	External1 Temp Offset	I1	–128	127	0	0.1°C
Calibration	Internal Temp Model	0x4400	Int Gain	I2	–32768	32767	–12143	—
Calibration	Internal Temp Model	0x4402	Int base offset	I2	–32768	32767	6232	—
Calibration	Internal Temp Model	0x4404	Int Minimum AD	I2	–32768	32767	0	—
Calibration	Internal Temp Model	0x4406	Int Maximum Temp	I2	–32768	32767	6232	0.1 K
Calibration	Cell Temperature Model	0x4408	Coeff a1	I2	–32768	32767	–11130	—
Calibration	Cell Temperature Model	0x440A	Coeff a2	I2	–32768	32767	19142	—
Calibration	Cell Temperature Model	0x440C	Coeff a3	I2	–32768	32767	–19262	—
Calibration	Cell Temperature Model	0x440E	Coeff a4	I2	–32768	32767	28203	—
Calibration	Cell Temperature Model	0x4410	Coeff a5	I2	–32768	32767	892	—
Calibration	Cell Temperature Model	0x4412	Coeff b1	I2	–32768	32767	328	—
Calibration	Cell Temperature Model	0x4414	Coeff b2	I2	–32768	32767	–605	—
Calibration	Cell Temperature Model	0x4416	Coeff b3	I2	–32768	32767	–2443	—
Calibration	Cell Temperature Model	0x4418	Coeff b4	I2	–32768	32767	4696	—
Calibration	Cell Temperature Model	0x441A	Rc0	I2	–32768	32767	11703	—
Calibration	Cell Temperature Model	0x441C	Adc0	I2	–32768	32767	11703	—

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Table 16-1. Data Flash Table (continued)

Class	Subclass	Address	Name	Type	Min Value	Max Value	Default	Units
Calibration	Cell Temperature Model	0x441E	Rpad	I2	–32768	32767	0	—
Calibration	Cell Temperature Model	0x4420	Rint	I2	–32768	32767	0	—
Calibration	Current Deadband	0x4446	Deadband	U1	0	255	3	mA
Calibration	Current Deadband	0x4447	Coulomb Counter Deadband	U1	0	255	9	116 nV
Settings	Protection	0x46AD	Protection Configuration	H1	0x0	0x03	0x0	Hex
Settings	Protection	0x46AE	Enabled Protections A	H1	0x0	0xFF	0x57	Hex
Settings	Protection	0x46AF	Enabled Protections B	H1	0x0	0xFF	0x35	Hex
Settings	Protection	0x46B0	Enabled Protections C	H1	0x0	0xFF	0x14	Hex
Settings	Protection	0x46B1	Enabled Protections D	H1	0x0	0xFF	0x0C	Hex
Settings	Permanent Failure	0x46FB	Enabled PF A	H1	0x0	0xFF	0x0	Hex
Settings	Permanent Failure	0x46FC	Enabled PF B	H1	0x0	0xFF	0x0	Hex
Settings	Permanent Failure	0x46FD	Enabled PF C	H1	0x0	0xFF	0x0	Hex
Settings	Configuration	0x4600	FET Options	H1	0x0	0xFF	0x20	Hex
Settings	Configuration	0x4601	I2C Gauging Configuration	H1	0x0	0x0F	0x04	Hex
Settings	Configuration	0x4602	I2C Configuration	H1	0x0	0xFF	0x01	Hex
Settings	Configuration	0x4604	Power Config	H1	0x0	0xBF	0x00	Hex
Settings	Configuration	0x4631	SOC Flag Config A	H2	0x0	0x0FFF	0x0C8C	Hex
Settings	Configuration	0x4633	SOC Flag Config B	H1	0x0	0xFF	0x8C	Hex
Settings	Configuration	0x464C	IT Gauging Configuration	H2	0x0	0xFFFF	0x14CE	Hex
Settings	Configuration	0x465D	Charging Configuration	H1	0x0	0x3F	0x0	Hex
Settings	Configuration	0x4699	Temperature Enable	H1	0x0	0x03	0x03	Hex
Settings	Configuration	0x469A	DA Configuration	H1	0x0	0xFF	0x11	Hex
Settings	Configuration	0x470C	Balancing Configuration	H1	0x0	0xFF	0x01	Hex
Settings	AFE	0x469C	AFE Protection Control	H1	0x0	0xFF	0x70	Hex
Settings	AFE	0x46A2	ZVCHG Exit Threshold	I2	0	8000	2200	mV
Settings	Charger	0x4623	Device Address	H1	0x0	0xFF	0xD4	Hex
Settings	Charger	0x4624	Voltage Register	H1	0x0	0xFF	0x0C	Hex
Settings	Charger	0x4625	Current Register	H1	0x0	0xFF	0x0A	Hex
Settings	Charger	0x4628	Broadcast Pacing	U1	0	255	15	s
Settings	Manufacturing	0x43C0	Mfg Status init	H2	0x0	0xFFFF	0x0000	Hex
Protections	CUV	0x46B2	Threshold	I2	0	32767	2500	mV
Protections	CUV	0x46B4	Delay	U1	0	255	2	s
Protections	CUV	0x46B5	Recovery	I2	0	32767	3000	mV
Protections	COV	0x46B7	Threshold Low Temp	I2	0	32767	4300	mV
Protections	COV	0x46B9	Threshold Standard Temp	I2	0	32767	4300	mV
Protections	COV	0x46BB	Threshold High Temp	I2	0	32767	4300	mV
Protections	COV	0x46BD	Threshold Rec Temp	I2	0	32767	4300	mV
Protections	COV	0x46BF	Delay	U1	0	255	2	s
Protections	COV	0x46C0	Recovery Low Temp	I2	0	32767	3900	mV
Protections	COV	0x46C2	Recovery Standard Temp	I2	0	32767	3900	mV
Protections	COV	0x46C4	Recovery High Temp	I2	0	32767	3900	mV
Protections	COV	0x46C6	Recovery Rec Temp	I2	0	32767	3900	mV
Protections	OCC	0x46C8	Threshold	I2	–32768	32767	6000	mA
Protections	OCC	0x46CA	Delay	U1	0	255	6	s
Protections	OCC	0x46CB	Recovery Threshold	I2	–32768	32767	–200	mA
Protections	OCC	0x46CD	Recovery Delay	U1	0	255	5	s
Protections	OCD	0x46CE	Threshold	I2	–32768	32767	–6000	mA
Protections	OCD	0x46D0	Delay	U1	0	255	6	s

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Table 16-1. Data Flash Table (continued)

Class	Subclass	Address	Name	Type	Min Value	Max Value	Default	Units
Protections	OCD	0x46D1	Recovery Threshold	I2	–32768	32767	200	mA
Protections	OCD	0x46D3	Recovery Delay	U1	0	255	5	s
Protections	AOLD	0x469D	Threshold	H1	0x0	0xFF	0xF4	Hex
Protections	AOLD	0x46D4	Recovery	U1	0	255	5	s
Protections	ASCC	0x469E	Threshold	H1	0x0	0xFF	0x77	Hex
Protections	ASCC	0x46D5	Recovery	U1	0	255	5	s
Protections	ASCD	0x469F	Threshold 1	H1	0x0	0xFF	0x77	Hex
Protections	ASCD	0x46A0	Threshold 2	H1	0x0	0xFF	0xE7	Hex
Protections	ASCD	0x46D6	Recovery	U1	0	255	5	s
Protections	OTC	0x46D7	Threshold	I2	–400	1500	550	0.1°C
Protections	OTC	0x46D9	Delay	U1	0	255	2	s
Protections	OTC	0x46DA	Recovery	I2	–400	1500	500	0.1°C
Protections	OTD	0x46DC	Threshold	I2	–400	1500	600	0.1°C
Protections	OTD	0x46DE	Delay	U1	0	255	2	s
Protections	OTD	0x46DF	Recovery	I2	–400	1500	550	0.1°C
Protections	UTC	0x46E1	Threshold	I2	–400	1500	0	0.1°C
Protections	UTC	0x46E3	Delay	U1	0	255	2	s
Protections	UTC	0x46E4	Recovery	I2	–400	1500	50	0.1°C
Protections	UTD	0x46E6	Threshold	I2	–400	1500	0	0.1°C
Protections	UTD	0x46E8	Delay	U1	0	255	2	s
Protections	UTD	0x46E9	Recovery	I2	–400	1500	50	0.1°C
Protections	PTO	0x46EB	Charge Threshold	I2	–32768	32767	2000	mA
Protections	PTO	0x46ED	Suspend Threshold	I2	–32768	32767	1800	mA
Protections	PTO	0x46EF	Delay	U2	0	65535	1800	s
Protections	PTO	0x46F1	Reset	I2	0	32767	2	mAh
Protections	CTO	0x46F3	Charge Threshold	I2	–32768	32767	2500	mA
Protections	CTO	0x46F5	Suspend Threshold	I2	–32768	32767	2000	mA
Protections	CTO	0x46F7	Delay	U2	0	65535	54000	s
Protections	CTO	0x46F9	Reset	I2	0	32767	2	mAh
Permanent Fail	SUV	0x46FF	Threshold	I2	0	32767	1000	mV
Permanent Fail	SUV	0x4701	Delay	U1	0	255	5	s
Permanent Fail	SOV	0x4702	Threshold	I2	0	32767	4500	mV
Permanent Fail	SOV	0x4704	Delay	U1	0	255	5	s
Permanent Fail	CFET	0x4705	OFF Threshold	I2	0	500	5	mA
Permanent Fail	CFET	0x4707	OFF Delay	U1	0	255	5	s
Permanent Fail	DFET	0x4708	OFF Threshold	I2	–500	0	–5	mA
Permanent Fail	DFET	0x470A	OFF Delay	U1	0	255	5	s
Permanent Fail	VIMR	0x4717	Check Voltage	I2	0	5000	3500	mV
Permanent Fail	VIMR	0x4719	Check Current	I2	0	32767	10	mA
Permanent Fail	VIMR	0x471B	Delta Threshold	I2	0	5000	500	mV
Permanent Fail	VIMR	0x471D	Delta Delay	U1	0	255	5	s
Permanent Fail	VIMR	0x471E	Duration	U2	0	65535	100	s
Permanent Fail	VIMA	0x4720	Check Voltage	I2	0	5000	3700	mV
Permanent Fail	VIMA	0x4722	Check Current	I2	0	32767	50	mA
Permanent Fail	VIMA	0x4724	Delta Threshold	I2	0	5000	200	mV
Permanent Fail	VIMA	0x4726	Delay	U1	0	255	5	s
Advanced Charge Algorithm	Temperature Ranges	0x465E	T1 Temp	I1	–128	127	0	°C
Advanced Charge Algorithm	Temperature Ranges	0x465F	T2 Temp	I1	–128	127	12	°C

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Table 16-1. Data Flash Table (continued)

Class	Subclass	Address	Name	Type	Min Value	Max Value	Default	Units
Advanced Charge Algorithm	Temperature Ranges	0x4660	T5 Temp	I1	–128	127	20	°C
Advanced Charge Algorithm	Temperature Ranges	0x4661	T6 Temp	I1	–128	127	25	°C
Advanced Charge Algorithm	Temperature Ranges	0x4662	T3 Temp	I1	–128	127	30	°C
Advanced Charge Algorithm	Temperature Ranges	0x4663	T4 Temp	I1	–128	127	55	°C
Advanced Charge Algorithm	Temperature Ranges	0x4664	Hysteresis Temp	I1	0	127	1	°C
Advanced Charge Algorithm	Low Temp Charging	0x4665	Voltage	I2	0	32767	4000	mV
Advanced Charge Algorithm	Low Temp Charging	0x4667	Current Low	I2	0	32767	132	mA
Advanced Charge Algorithm	Low Temp Charging	0x4669	Current Med	I2	0	32767	352	mA
Advanced Charge Algorithm	Low Temp Charging	0x466B	Current High	I2	0	32767	264	mA
Advanced Charge Algorithm	Standard Temp Charging	0x466D	Voltage	I2	0	32767	4200	mV
Advanced Charge Algorithm	Standard Temp Charging	0x466F	Current Low	I2	0	32767	1980	mA
Advanced Charge Algorithm	Standard Temp Charging	0x4671	Current Med	I2	0	32767	4004	mA
Advanced Charge Algorithm	Standard Temp Charging	0x4673	Current High	I2	0	32767	2992	mA
Advanced Charge Algorithm	High Temp Charging	0x4675	Voltage	I2	0	32767	4000	mV
Advanced Charge Algorithm	High Temp Charging	0x4677	Current Low	I2	0	32767	1012	mA
Advanced Charge Algorithm	High Temp Charging	0x4679	Current Med	I2	0	32767	1980	mA
Advanced Charge Algorithm	High Temp Charging	0x467B	Current High	I2	0	32767	1496	mA
Advanced Charge Algorithm	Rec Temp Charging	0x467D	Voltage	I2	0	32767	4100	mV
Advanced Charge Algorithm	Rec Temp Charging	0x467F	Current Low	I2	0	32767	2508	mA
Advanced Charge Algorithm	Rec Temp Charging	0x4681	Current Med	I2	0	32767	4488	mA
Advanced Charge Algorithm	Rec Temp Charging	0x4683	Current High	I2	0	32767	3520	mA
Advanced Charge Algorithm	Pre-Charging	0x4685	Current	I2	0	32767	88	mA
Advanced Charge Algorithm	Maintenance Charging	0x4687	Current	I2	0	32767	44	mA
Advanced Charge Algorithm	Voltage Range	0x4689	Precharge Start Voltage	I2	0	32767	2500	mV
Advanced Charge Algorithm	Voltage Range	0x468B	Charging Voltage Low	I2	0	32767	2900	mV
Advanced Charge Algorithm	Voltage Range	0x468D	Charging Voltage Med	I2	0	32767	3600	mV
Advanced Charge Algorithm	Voltage Range	0x468F	Charging Voltage High	I2	0	32767	4000	mV
Advanced Charge Algorithm	Voltage Range	0x4691	Charging Voltage Hysteresis	U1	0	255	0	mV
Advanced Charge Algorithm	Termination Config	0x4692	Charge Term Taper Current	I2	0	32767	250	mA
Advanced Charge Algorithm	Termination Config	0x4696	Charge Term Voltage	I2	0	32767	75	mV

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Table 16-1. Data Flash Table (continued)

Class	Subclass	Address	Name	Type	Min Value	Max Value	Default	Units
Advanced Charge Algorithm	Cell Balancing Config	0x470D	Bal Time/mAh Cell 1	U2	0	65535	354	s/mAh
Advanced Charge Algorithm	Cell Balancing Config	0x470F	Bal Time/mAh Cell 2	U2	0	65535	361	s/mAh
Advanced Charge Algorithm	Cell Balancing Config	0x4711	Min Start Balance Delta	U1	0	255	3	mV
Advanced Charge Algorithm	Cell Balancing Config	0x4712	Relax Balance Interval	U4	0	4294967295	18000	s
Advanced Charge Algorithm	Cell Balancing Config	0x4716	Min Rsoc for Balancing	U1	0	100	80	%
Gas Gauging	Current Thresholds	0x46A5	Dsg Current Threshold	I2	–32768	32767	100	mA
Gas Gauging	Current Thresholds	0x46A7	Chg Current Threshold	I2	–32768	32767	50	mA
Gas Gauging	Current Thresholds	0x46A9	Quit Current	I2	0	32767	10	mA
Gas Gauging	Current Thresholds	0x46AB	Dsg Relax Time	U1	0	255	1	s
Gas Gauging	Current Thresholds	0x46AC	Chg Relax Time	U1	0	255	60	s
Gas Gauging	Standby	0x4618	StandbyCurrent	I2	–32768	0	–10	mA
Gas Gauging	Max Load	0x461A	Max Load Current	I2	–32768	0	–500	mA
Gas Gauging	Max Load	0x461C	Max Load Rsoc	U1	0	100	50	%
Gas Gauging	Design	0x4629	Design Capacity mAh	I2	0	32767	4400	mAh
Gas Gauging	Design	0x462B	Design Capacity cWh	I2	0	32767	6336	cWh
Gas Gauging	Design	0x462D	Design Voltage	I2	0	32767	7200	mV
Gas Gauging	Cycle	0x462F	Cycle Count Percentage	U1	0	100	90	%
Gas Gauging	FD	0x4634	Set Voltage Threshold	I2	0	5000	3000	mV
Gas Gauging	FD	0x4636	Clear Voltage Threshold	I2	0	5000	3100	mV
Gas Gauging	FD	0x4638	Set % RSOC Threshold	U1	0	100	0	%
Gas Gauging	FD	0x4639	Clear % RSOC Threshold	U1	0	100	5	%
Gas Gauging	FC	0x463A	Set Voltage Threshold	I2	0	5000	4200	mV
Gas Gauging	FC	0x463C	Clear Voltage Threshold	I2	0	5000	4100	mV
Gas Gauging	FC	0x463E	Set % RSOC Threshold	U1	0	100	100	%
Gas Gauging	FC	0x463F	Clear % RSOC Threshold	U1	0	100	95	%
Gas Gauging	TD	0x4640	Set Voltage Threshold	I2	0	5000	3200	mV
Gas Gauging	TD	0x4642	Clear Voltage Threshold	I2	0	5000	3300	mV
Gas Gauging	TD	0x4644	Set % RSOC Threshold	U1	0	100	6	%
Gas Gauging	TD	0x4645	Clear % RSOC Threshold	U1	0	100	8	%
Gas Gauging	TC	0x4646	Set Voltage Threshold	I2	0	5000	4200	mV
Gas Gauging	TC	0x4648	Clear Voltage Threshold	I2	0	5000	4100	mV
Gas Gauging	TC	0x464A	Set % RSOC Threshold	U1	0	100	100	%
Gas Gauging	TC	0x464B	Clear % RSOC Threshold	U1	0	100	95	%
Gas Gauging	State	0x4206	Qmax Cell 1	I2	0	32767	4400	mAh
Gas Gauging	State	0x4208	Qmax Cell 2	I2	0	32767	4400	mAh
Gas Gauging	State	0x420A	Qmax Pack	I2	0	32767	4400	mAh
Gas Gauging	State	0x420C	Qmax Cycle Count	U2	0	65535	0	—
Gas Gauging	State	0x420E	Update Status	H1	0x0	0x0E	0x0	—
Gas Gauging	State	0x420F	Cell 1 Chg Voltage at EoC	I2	0	32767	4200	mV
Gas Gauging	State	0x4211	Cell 2 Chg Voltage at EoC	I2	0	32767	4200	mV
Gas Gauging	State	0x4213	Current at EoC	I2	0	32767	250	mA
Gas Gauging	State	0x4215	Avg I Last Run	I2	–32768	32767	–2000	mA
Gas Gauging	State	0x4217	Avg P Last Run	I2	–32768	32767	–3022	cW
Gas Gauging	State	0x4219	Delta Voltage	I2	–32768	32767	0	mV
Gas Gauging	State	0x421B	Temp k	I2	0	32767	100	0.1°C/256 cW

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Table 16-1. Data Flash Table (continued)

Class	Subclass	Address	Name	Type	Min Value	Max Value	Default	Units
Gas Gauging	State	0x421D	Temp a	I2	0	32767	1000	s
Gas Gauging	State	0x421F	Max Avg I Last Run	I2	–32768	32767	–2000	mA
Gas Gauging	State	0x4221	Max Avg P Last Run	I2	–32768	32767	–3022	cW
Gas Gauging	State	0x4240	Cycle Count	U2	0	65535	0	—
Gas Gauging	State	0x4727	SOH Temp k	I2	0	32767	100	0.1°C/256 cW
Gas Gauging	State	0x4729	SOH Temp a	I2	0	32767	1000	s
Gas Gauging	IT Cfg	0x4200	Design Resistance	I2	1	32767	96	mΩ
Gas Gauging	IT Cfg	0x4202	Pack Resistance	I2	0	32767	30	mΩ
Gas Gauging	IT Cfg	0x4204	System Resistance	I2	0	32767	0	mΩ
Gas Gauging	IT Cfg	0x458E	Ra Filter	U2	0	999	800	0.1%
Gas Gauging	IT Cfg	0x4591	Ra Max Delta	U1	0	255	15	%
Gas Gauging	IT Cfg	0x4594	Resistance Parameter Filter	U2	1	65535	65142	—
Gas Gauging	IT Cfg	0x4596	Near EDV Ra Param Filter	U2	1	65535	59220	—
Gas Gauging	IT Cfg	0x45BC	Qmax Delta	U1	3	100	5	%
Gas Gauging	IT Cfg	0x45BD	Qmax Upper Bound	U1	100	255	130	%
Gas Gauging	IT Cfg	0x45BE	Term Voltage	I2	0	32767	6000	mV
Gas Gauging	IT Cfg	0x45C0	Term V Hold Time	U1	0	255	1	s
Gas Gauging	IT Cfg	0x45C1	Term Voltage Delta	I2	0	32767	300	mV
Gas Gauging	IT Cfg	0x45C3	Term Min Cell V	I2	0	32767	2800	mV
Gas Gauging	IT Cfg	0x45CC	Max Simulation Iterations	U1	20	50	30	—
Gas Gauging	IT Cfg	0x45DD	Fast Scale Start SOC	U1	0	100	10	%
Gas Gauging	IT Cfg	0x45E6	Min Delta Voltage	I2	–32768	32767	0	mV
Gas Gauging	IT Cfg	0x4650	Load Select	U1	0	7	7	—
Gas Gauging	IT Cfg	0x4651	Load Mode	U1	0	1	0	—
Gas Gauging	IT Cfg	0x4652	User Rate-mA	I2	–9000	0	0	mA
Gas Gauging	IT Cfg	0x4654	User Rate-cW	I2	–32768	0	0	cW
Gas Gauging	IT Cfg	0x4656	Reserve Cap-mAh	I2	0	9000	0	mAh
Gas Gauging	IT Cfg	0x4658	Reserve Cap-cWh	I2	0	32000	0	cWh
Gas Gauging	Smoothing	0x465A	Smooth Relax Time	U2	1	32767	1000	s
Gas Gauging	Condition Flag	0x465C	Max Error Limit	U1	0	100	100	%
Gas Gauging	Max Error	0x472E	Time Cycle Equivalent	U1	1	255	12	2 h
Gas Gauging	Max Error	0x472F	Cycle Delta	U1	0	255	5	0.01%
Gas Gauging	SoH	0x45E8	SoH Load Rate	U1	0	255	50	0.1 Hr rate
Gas Gauging	SoH	0x45E9	SoH Taper Current	I2	0	32767	250	mA
Gas Gauging	SoH	0x45EB	SoH Charge Voltage	I2	0	32767	4420	mV
Power	Power	0x4605	Valid Update Voltage	I2	0	32767	2800	mV
Power	Shutdown	0x4607	Shutdown Voltage	I2	0	32767	2300	mV
Power	Shutdown	0x4609	Shutdown Time	U1	0	255	10	s
Power	Shutdown	0x460A	Charger Present Threshold	I2	0	32767	3000	mV
Power	Sleep	0x460C	Sleep Current	I2	0	32767	10	mA
Power	Sleep	0x460E	Bus Timeout	U1	0	255	5	s
Power	Sleep	0x4613	Voltage Time	U1	1	20	5	s
Power	Sleep	0x4614	Current Time	U1	1	60	20	s
Power	Sleep	0x4615	Wake Comparator	H1	0x0	0xFF	0x0	Hex
Power	Ship	0x4616	FET Off Time	U1	0	127	10	s
Power	Ship	0x4617	Delay	U1	0	254	20	s
Power	Ship	0x461D	Auto Ship Time	U2	0	6553	144	10 min
Power	PMPDRV	0x461F	PMPDRV On VC2	I2	0	32767	3000	mV
PF Status	Device Status Data	0x42C0	Safety Alert A	H1	0x0	0xFF	0x0	Hex

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Table 16-1. Data Flash Table (continued)

Class	Subclass	Address	Name	Type	Min Value	Max Value	Default	Units
PF Status	Device Status Data	0x42C1	Safety Status A	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42C2	Safety Alert B	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42C3	Safety Status B	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42C4	Safety Alert C	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42C5	Safety Status C	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42C6	Safety Alert D	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42C7	Safety Status D	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42C8	PF Alert A	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42C9	PF Status A	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42CA	PF Alert B	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42CB	PF Status B	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42CC	PF Alert C	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42CD	PF Status C	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42CF	PF Status D	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42D0	Operation Status A	H2	0x0	0xFFFF	0x0	Hex
PF Status	Device Status Data	0x42D2	Operation Status B	H2	0x0	0xFFFF	0x0	Hex
PF Status	Device Status Data	0x42D4	Temp Range	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42D5	Charging Status A	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42D7	Gauging Status	H1	0x0	0xFF	0x0	Hex
PF Status	Device Status Data	0x42D8	IT Status	H2	0x0	0xFFFF	0x0	Hex
PF Status	Device Voltage Data	0x42DA	Cell 1 Voltage	I2	–32768	32767	0	mV
PF Status	Device Voltage Data	0x42DC	Cell 2 Voltage	I2	–32768	32767	0	mV
PF Status	Device Voltage Data	0x42DE	Battery Direct Voltage	I2	–32768	32767	0	mV
PF Status	Device Voltage Data	0x42E0	Pack Voltage	I2	–32768	32767	0	mV
PF Status	Device Current Data	0x42E2	Current	I2	–32768	32767	0	mA
PF Status	Device Temperature Data	0x42E4	Internal Temperature	I2	–32768	32767	0	0.1°K
PF Status	Device Temperature Data	0x42E6	External 1 Temperature	I2	–32768	32767	0	0.1°K
PF Status	Device Gauging Data	0x42EA	Cell 1 Dod0	I2	–32768	32767	0	—
PF Status	Device Gauging Data	0x42EC	Cell 2 Dod0	I2	–32768	32767	0	—
PF Status	Device Gauging Data	0x42EE	Passed Charge	I2	–32768	32767	0	mAh
PF Status	AFE Regs	0x42F0	AFE Interrupt Status	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42F1	AFE FET Status	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42F2	AFE RXIN	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42F3	AFE Latch Status	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42F4	AFE Interrupt Enable	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42F5	AFE FET Control	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42F6	AFE RXIEN	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42F7	AFE RLOUT	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42F8	AFE RHOUT	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42F9	AFE RHINT	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42FA	AFE Cell Balance	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42FB	AFE AD/CC Control	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42FC	AFE ADC Mux	H1	0x0	0xFF	0x0	Hex

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Table 16-1. Data Flash Table (continued)

Class	Subclass	Address	Name	Type	Min Value	Max Value	Default	Units
PF Status	AFE Regs	0x42FD	AFE LED Output	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42FE	AFE State Control	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x42FF	AFE LED/Wake Control	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x4300	AFE Protection Control	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x4301	AFE OCD	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x4302	AFE SCC	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x4303	AFE SCD1	H1	0x0	0xFF	0x0	Hex
PF Status	AFE Regs	0x4304	AFE SCD2	H1	0x0	0xFF	0x0	Hex
System Data	Manufacturer Data	0x4041	Manufacturer Info Block A01	H1	0x0	0xFF	0x61	Hex
System Data	Manufacturer Data	0x4042	Manufacturer Info Block A02	H1	0x0	0xFF	0x62	Hex
System Data	Manufacturer Data	0x4043	Manufacturer Info Block A03	H1	0x0	0xFF	0x63	Hex
System Data	Manufacturer Data	0x4044	Manufacturer Info Block A04	H1	0x0	0xFF	0x64	Hex
System Data	Manufacturer Data	0x4045	Manufacturer Info Block A05	H1	0x0	0xFF	0x65	Hex
System Data	Manufacturer Data	0x4046	Manufacturer Info Block A06	H1	0x0	0xFF	0x66	Hex
System Data	Manufacturer Data	0x4047	Manufacturer Info Block A07	H1	0x0	0xFF	0x67	Hex
System Data	Manufacturer Data	0x4048	Manufacturer Info Block A08	H1	0x0	0xFF	0x68	Hex
System Data	Manufacturer Data	0x4049	Manufacturer Info Block A09	H1	0x0	0xFF	0x69	Hex
System Data	Manufacturer Data	0x404A	Manufacturer Info Block A10	H1	0x0	0xFF	0x6A	Hex
System Data	Manufacturer Data	0x404B	Manufacturer Info Block A11	H1	0x0	0xFF	0x6B	Hex
System Data	Manufacturer Data	0x404C	Manufacturer Info Block A12	H1	0x0	0xFF	0x6C	Hex
System Data	Manufacturer Data	0x404D	Manufacturer Info Block A13	H1	0x0	0xFF	0x6D	Hex
System Data	Manufacturer Data	0x404E	Manufacturer Info Block A14	H1	0x0	0xFF	0x6E	Hex
System Data	Manufacturer Data	0x404F	Manufacturer Info Block A15	H1	0x0	0xFF	0x6F	Hex
System Data	Manufacturer Data	0x4050	Manufacturer Info Block A16	H1	0x0	0xFF	0x70	Hex
System Data	Manufacturer Data	0x4051	Manufacturer Info Block A17	H1	0x0	0xFF	0x71	Hex
System Data	Manufacturer Data	0x4052	Manufacturer Info Block A18	H1	0x0	0xFF	0x72	Hex
System Data	Manufacturer Data	0x4053	Manufacturer Info Block A19	H1	0x0	0xFF	0x73	Hex
System Data	Manufacturer Data	0x4054	Manufacturer Info Block A20	H1	0x0	0xFF	0x74	Hex
System Data	Manufacturer Data	0x4055	Manufacturer Info Block A21	H1	0x0	0xFF	0x75	Hex
System Data	Manufacturer Data	0x4056	Manufacturer Info Block A22	H1	0x0	0xFF	0x76	Hex
System Data	Manufacturer Data	0x4057	Manufacturer Info Block A23	H1	0x0	0xFF	0x77	Hex
System Data	Manufacturer Data	0x4058	Manufacturer Info Block A24	H1	0x0	0xFF	0x7A	Hex
System Data	Manufacturer Data	0x4059	Manufacturer Info Block A25	H1	0x0	0xFF	0x78	Hex
System Data	Manufacturer Data	0x405A	Manufacturer Info Block A26	H1	0x0	0xFF	0x79	Hex
System Data	Manufacturer Data	0x405B	Manufacturer Info Block A27	H1	0x0	0xFF	0x30	Hex
System Data	Manufacturer Data	0x405C	Manufacturer Info Block A28	H1	0x0	0xFF	0x31	Hex
System Data	Manufacturer Data	0x405D	Manufacturer Info Block A29	H1	0x0	0xFF	0x32	Hex
System Data	Manufacturer Data	0x405E	Manufacturer Info Block A30	H1	0x0	0xFF	0x33	Hex
System Data	Manufacturer Data	0x405F	Manufacturer Info Block A31	H1	0x0	0xFF	0x34	Hex
System Data	Manufacturer Data	0x4060	Manufacturer Info Block A32	H1	0x0	0xFF	0x35	Hex
System Data	Integrity	0x4061	Static DF Signature	H2	0x0	0x7FFF	0x0	Hex
System Data	Integrity	0x4063	Static Chem DF Signature	H2	0x0	0x7FFF	0x6C98	Hex
System Data	Integrity	0x4065	All DF Signature	H2	0x0	0x7FFF	0x0	Hex
I2C Configuration	Data	0x4067	Manufacture Date	U2	0	65535	0	date
I2C Configuration	Data	0x4069	Serial Number	H2	0x0	0xFFFF	0x0001	Hex
I2C Configuration	Data	0x406B	Manufacturer Name	S21	x	x	Texas Instruments	—
I2C Configuration	Data	0x4080	Device Name	S21	x	x	bq28z620	—
I2C Configuration	Data	0x4095	Device Chemistry	S5	x	x	LION	—
Lifetimes	Voltage	0x4280	Cell 1 Max Voltage	I2	0	32767	0	mV

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Table 16-1. Data Flash Table (continued)

Class	Subclass	Address	Name	Type	Min Value	Max Value	Default	Units
Lifetimes	Voltage	0x4282	Cell 2 Max Voltage	I2	0	32767	0	mV
Lifetimes	Current	0x4284	Max Charge Current	I2	0	32767	0	mA
Lifetimes	Current	0x4286	Max Discharge Current	I2	–32768	0	0	mA
Lifetimes	Temperature	0x4288	Max Temp Cell	I1	–128	127	–128	°C
Lifetimes	Temperature	0x4289	Min Temp Cell	I1	–128	127	127	°C
Lifetimes	Safety Events	0x428A	No Of COV Events	U2	0	32767	0	events
Lifetimes	Safety Events	0x428C	Last COV Event	U2	0	32767	0	cycles
Lifetimes	Safety Events	0x428E	No Of CUV Events	U2	0	32767	0	events
Lifetimes	Safety Events	0x4290	Last CUV Event	U2	0	32767	0	cycles
Lifetimes	Safety Events	0x4292	No Of OCC1 Events	U2	0	32767	0	events
Lifetimes	Safety Events	0x4294	Last OCC1 Event	U2	0	32767	0	cycles
Lifetimes	Safety Events	0x4296	No Of ASCC Events	U2	0	32767	0	events
Lifetimes	Safety Events	0x4298	Last ASCC Event	U2	0	32767	0	cycles
Lifetimes	Power Events	0x429A	No Of Shutdowns	U1	0	255	0	events
Ra Table	R_a0	0x4100	Cell0 R_a flag	H2	0x0	0xFFFF	0xFF55	—
Ra Table	R_a0	0x4102	Cell0 R_a 0	I2	0	32767	67	2^{-10} Ω
Ra Table	R_a0	0x4104	Cell0 R_a 1	I2	0	32767	71	2^{-10} Ω
Ra Table	R_a0	0x4106	Cell0 R_a 2	I2	0	32767	83	2^{-10} Ω
Ra Table	R_a0	0x4108	Cell0 R_a 3	I2	0	32767	110	2^{-10} Ω
Ra Table	R_a0	0x410A	Cell0 R_a 4	I2	0	32767	96	2^{-10} Ω
Ra Table	R_a0	0x410C	Cell0 R_a 5	I2	0	32767	77	2^{-10} Ω
Ra Table	R_a0	0x410E	Cell0 R_a 6	I2	0	32767	96	2^{-10} Ω
Ra Table	R_a0	0x4110	Cell0 R_a 7	I2	0	32767	86	2^{-10} Ω
Ra Table	R_a0	0x4112	Cell0 R_a 8	I2	0	32767	84	2^{-10} Ω
Ra Table	R_a0	0x4114	Cell0 R_a 9	I2	0	32767	82	2^{-10} Ω
Ra Table	R_a0	0x4116	Cell0 R_a 10	I2	0	32767	81	2^{-10} Ω
Ra Table	R_a0	0x4118	Cell0 R_a 11	I2	0	32767	92	2^{-10} Ω
Ra Table	R_a0	0x411A	Cell0 R_a 12	I2	0	32767	103	2^{-10} Ω
Ra Table	R_a0	0x411C	Cell0 R_a 13	I2	0	32767	123	2^{-10} Ω
Ra Table	R_a0	0x411E	Cell0 R_a 14	I2	0	32767	658	2^{-10} Ω
Ra Table	R_a1	0x4140	Cell1 R_a flag	H2	0x0	0xFFFF	0xFF55	—
Ra Table	R_a1	0x4142	Cell1 R_a 0	I2	0	32767	67	2^{-10} Ω
Ra Table	R_a1	0x4144	Cell1 R_a 1	I2	0	32767	71	2^{-10} Ω
Ra Table	R_a1	0x4146	Cell1 R_a 2	I2	0	32767	83	2^{-10} Ω
Ra Table	R_a1	0x4148	Cell1 R_a 3	I2	0	32767	110	2^{-10} Ω
Ra Table	R_a1	0x414A	Cell1 R_a 4	I2	0	32767	96	2^{-10} Ω
Ra Table	R_a1	0x414C	Cell1 R_a 5	I2	0	32767	77	2^{-10} Ω
Ra Table	R_a1	0x414E	Cell1 R_a 6	I2	0	32767	96	2^{-10} Ω
Ra Table	R_a1	0x4150	Cell1 R_a 7	I2	0	32767	86	2^{-10} Ω
Ra Table	R_a1	0x4152	Cell1 R_a 8	I2	0	32767	84	2^{-10} Ω
Ra Table	R_a1	0x4154	Cell1 R_a 9	I2	0	32767	82	2^{-10} Ω
Ra Table	R_a1	0x4156	Cell1 R_a 10	I2	0	32767	81	2^{-10} Ω
Ra Table	R_a1	0x4158	Cell1 R_a 11	I2	0	32767	92	2^{-10} Ω
Ra Table	R_a1	0x415A	Cell1 R_a 12	I2	0	32767	103	2^{-10} Ω
Ra Table	R_a1	0x415C	Cell1 R_a 13	I2	0	32767	123	2^{-10} Ω
Ra Table	R_a1	0x415E	Cell1 R_a 14	I2	0	32767	658	2^{-10} Ω
Ra Table	R_a0x	0x4180	xCell0 R_a flag	H2	0x0	0xFFFF	0xFFFF	—
Ra Table	R_a0x	0x4182	xCell0 R_a 0	I2	0	32767	67	2^{-10} Ω
Ra Table	R_a0x	0x4184	xCell0 R_a 1	I2	0	32767	71	2^{-10} Ω

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Table 16-1. Data Flash Table (continued)

Class	Subclass	Address	Name	Type	Min Value	Max Value	Default	Units
Ra Table	R_a0x	0x4186	xCell0 R_a 2	I2	0	32767	83	2^{-10} Ω
Ra Table	R_a0x	0x4188	xCell0 R_a 3	I2	0	32767	110	2^{-10} Ω
Ra Table	R_a0x	0x418A	xCell0 R_a 4	I2	0	32767	96	2^{-10} Ω
Ra Table	R_a0x	0x418C	xCell0 R_a 5	I2	0	32767	77	2^{-10} Ω
Ra Table	R_a0x	0x418E	xCell0 R_a 6	I2	0	32767	96	2^{-10} Ω
Ra Table	R_a0x	0x4190	xCell0 R_a 7	I2	0	32767	86	2^{-10} Ω
Ra Table	R_a0x	0x4192	xCell0 R_a 8	I2	0	32767	84	2^{-10} Ω
Ra Table	R_a0x	0x4194	xCell0 R_a 9	I2	0	32767	82	2^{-10} Ω
Ra Table	R_a0x	0x4196	xCell0 R_a 10	I2	0	32767	81	2^{-10} Ω
Ra Table	R_a0x	0x4198	xCell0 R_a 11	I2	0	32767	92	2^{-10} Ω
Ra Table	R_a0x	0x419A	xCell0 R_a 12	I2	0	32767	103	2^{-10} Ω
Ra Table	R_a0x	0x419C	xCell0 R_a 13	I2	0	32767	123	2^{-10} Ω
Ra Table	R_a0x	0x419E	xCell0 R_a 14	I2	0	32767	658	2^{-10} Ω
Ra Table	R_a1x	0x41C0	xCell1 R_a flag	H2	0x0	0xFFFF	0xFFFF	—
Ra Table	R_a1x	0x41C2	xCell1 R_a 0	I2	0	32767	67	2^{-10} Ω
Ra Table	R_a1x	0x41C4	xCell1 R_a 1	I2	0	32767	71	2^{-10} Ω
Ra Table	R_a1x	0x41C6	xCell1 R_a 2	I2	0	32767	83	2^{-10} Ω
Ra Table	R_a1x	0x41C8	xCell1 R_a 3	I2	0	32767	110	2^{-10} Ω
Ra Table	R_a1x	0x41CA	xCell1 R_a 4	I2	0	32767	96	2^{-10} Ω
Ra Table	R_a1x	0x41CC	xCell1 R_a 5	I2	0	32767	77	2^{-10} Ω
Ra Table	R_a1x	0x41CE	xCell1 R_a 6	I2	0	32767	96	2^{-10} Ω
Ra Table	R_a1x	0x41D0	xCell1 R_a 7	I2	0	32767	86	2^{-10} Ω
Ra Table	R_a1x	0x41D2	xCell1 R_a 8	I2	0	32767	84	2^{-10} Ω
Ra Table	R_a1x	0x41D4	xCell1 R_a 9	I2	0	32767	82	2^{-10} Ω
Ra Table	R_a1x	0x41D6	xCell1 R_a 10	I2	0	32767	81	2^{-10} Ω
Ra Table	R_a1x	0x41D8	xCell1 R_a 11	I2	0	32767	92	2^{-10} Ω
Ra Table	R_a1x	0x41DA	xCell1 R_a 12	I2	0	32767	103	2^{-10} Ω
Ra Table	R_a1x	0x41DC	xCell1 R_a 13	I2	0	32767	123	2^{-10} Ω
Ra Table	R_a1x	0x41DE	xCell1 R_a 14	I2	0	32767	658	2^{-10} Ω

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